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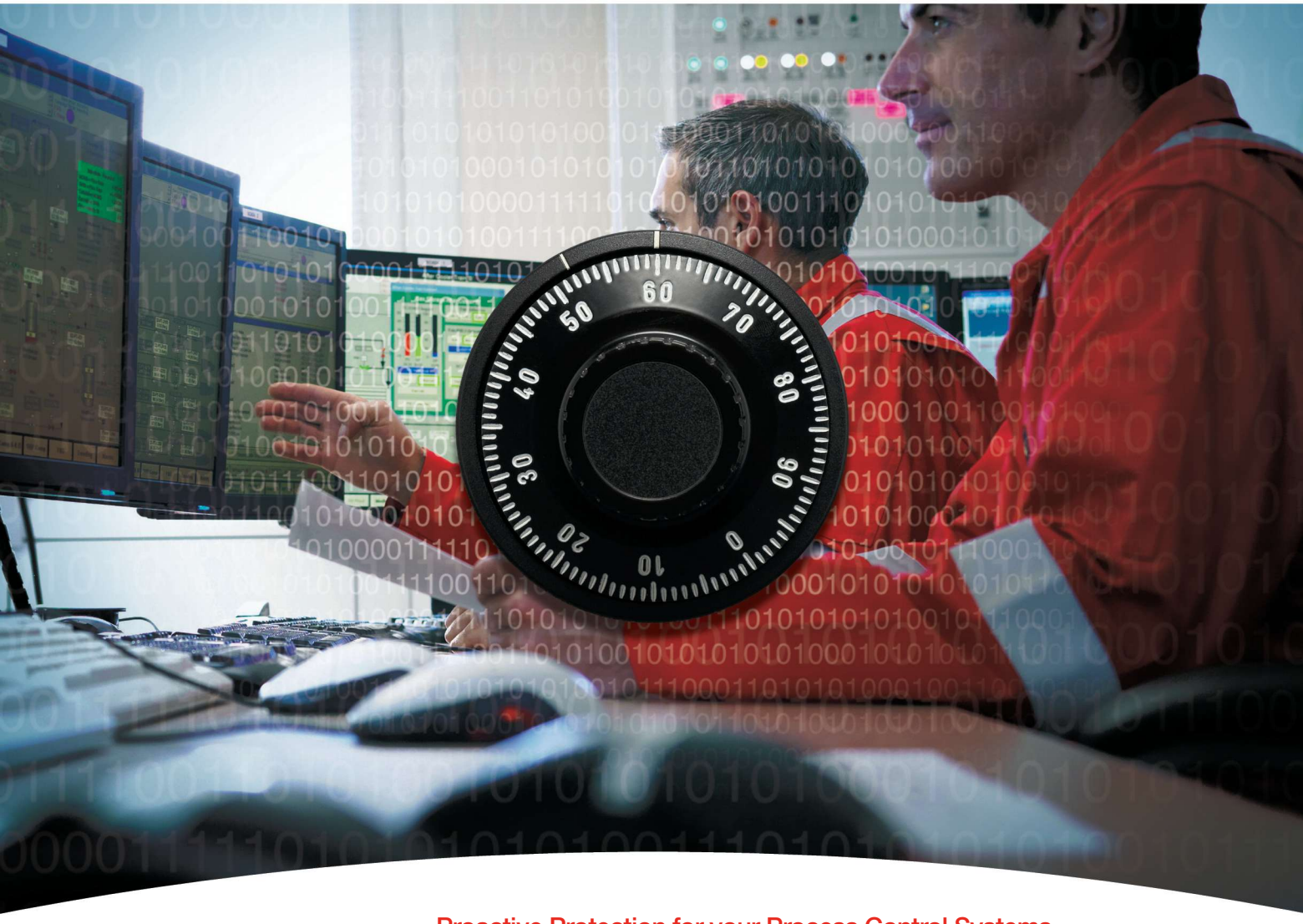
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COVER STORY

Next frontier: Operator-automation relationship

By Eddie Habibi

Industrial control systems are complex designs with little consideration to the needs of the human operator, which has caused serious and costly accidents. By empowering operators with the right human-automation relationship tools, a company will achieve and exceed goals.

PROCESS AUTOMATION

18 Big changes ahead for Fieldbus Foundation

By Larry O'Brien

The Foundation Fieldbus spec is evolving with an increased focus on usability and cross-industry cooperation. The big discussion lately is the memorandum of understanding about a potential merger of Fieldbus Foundation and HART Communication Foundation. The overall goal is to help users manage their intelligent devices effectively and get the maximum benefit.

FACTORY AUTOMATION

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By Dick Hill

Justifying an automation project requires a view of the big picture. Every change needs to make your company more competitive, in both the short and long term. Use that argument for justifying your project, and back it up with hard figures.

SYSTEM INTEGRATION

30 Field wireless networks

By Amit Ajmeri

Industrial wireless has specific requirements for performance, reliability, and security. Various wireless technologies, including the ISA-100.11a protocol, are employed to meet these requirements in areas such as topology, security, and customization.

AUTOMATION IT

36 OPC Foundation open standards

By Thomas J. Burke

OPC became the de facto standard for data acquisition for discrete and process control applications, and has evolved and revolutionized industrial automation multiplatform interoperability. The new OPC UA standard moves to the next level, enabling frictionless information integration.

SPECIAL SECTION: PRESSURE MEASUREMENT

40 Reference recorders shake up old standards

By David K. Porter, P.E.

Continuous innovation has become the norm within the pressure measurement industry. Commonly used, decades-old equipment is being phased out by modern instruments. And today, reference recorders are shaking up old standards in pressure measurement.

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WEB EXCLUSIVE

The top four criteria for manufacturing software comparison

This survey explores criteria for selecting manufacturing operations management software applications and identifies the four criteria users consider most important.

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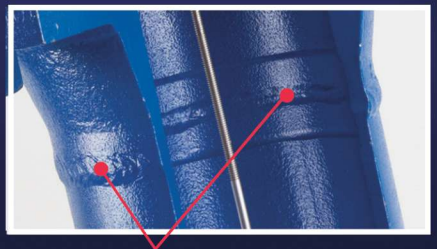
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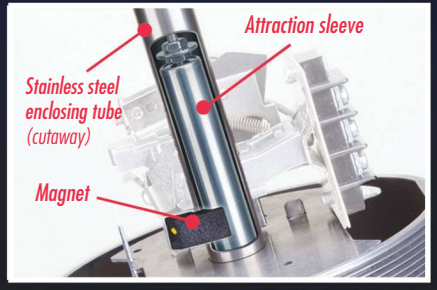
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Who done it?

By Bill Lydon, *InTech*, Chief Editor



Sometimes an old murder mystery novel or movie is referred to as a "who done it," and when we have issues in automation, manufacturing, and production, we have the real-world equivalent. The plot of these novels and movies was for the good guys to find out who performed the murder. Throughout the movie the good guys uncover information and use these clues and their intuition to solve the mystery. The most powerful tools the good guys use are observation, questions, and validating information. These are good tools we all can use to understand who or what "done it" when faced with automation, manufacturing, and production problems and issues.

The lesson is we need to be careful about the assumptions we make consciously and unconsciously when working on problems.

It is easy to jump to conclusions based on past experience or what obviously looks to be the problem. Many times at this point, it is good to step back and observe without judgment and ask, what is happening? As the great baseball coach Yogi Berra once stated, "You can observe a lot by just watching." The most basic level of observation is the gathering of information by using our five senses: sight, smell, hearing, taste, and touch. Based on this information, we make qualitative and quantitative observations. As technical people, we tend to like quantitative information that is absolute. But both types of observations are valuable, and exploring qualitative observations can lead to uncovering information that is quantitative. An operator, for example, can tell us a production process is not running properly; this is generally a qualitative observation. We can then start looking at process data, running trend reports, and quantifying what is happening.

When problems are particularly difficult, we can make inferences that help explain an observation we have made. The inferences are based on past experiences and prior knowledge that help illustrate what is happening. Inferences are valuable, but in unusual situations, they become a velvet trap leading us to the wrong conclusions. Many times, we are strongly influenced by the last problem we solved and assume this one is similar. This is where it is valuable and instructive to validate information as we uncover it. I recall, for example, spending a great deal of time troubleshooting an automation problem and essentially going around in circles troubleshooting control loops. As it turned out, I was relying on sensor readings that should have been operating properly and accurately. Unfortunately, there was an issue with a sensor interface, resulting in bad measured data. After a considerable amount of time troubleshooting, I desperately started checking all the system components and identified the faulty sensor interface. The lesson is we need to be careful about the assumptions we make consciously and unconsciously when working on problems.

The popular fictional detective Sherlock Holmes' companion, Dr. Watson, commented on a case, "This is indeed a mystery; what do you imagine it means?" Holmes replied, "I have no data yet. It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts." In some situations, we need to really drill down by asking questions to get facts.

We can all benefit from listening, observing, questioning, and validating information.

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Controller selection



Your May/June 2013 article "Considerations for selecting a controller – or server-based batch sequencer" makes good points about how legacy approaches to batch automation are now challenged to provide sophisticated control capabilities that enable increased throughput, lower costs, and improved regulatory compliance.

The article positions controller-based systems as being more appropriate for less-complex operations that require fewer upfront investments, while the server-based approach is heralded as the Cadillac solution. That may have been true many years ago, but the article does not acknowledge the recent advancements in controller-based batch systems.

The latest batch automation solutions are designed to manage all batch operations entirely in the controller. No complicated assessment is necessary. This approach provides a single operating and engineering platform for batch execution while meeting the objective of ISA-88, of separating engineering activities from recipe management. It also reduces complexity and improves availability and speed of execution. An operator can create a batch and run it to completion without dependence on a server anywhere in the system. The absence of a server removes associated licensing costs, security concerns and maintenance issues, as well as problems and production loss arising from communication latency.

Other benefits:

- The operator's view of the batch comes directly from the controller and is not dependent on a batch server. Operators gain access to standard displays for all batch functions.
- Plant availability is improved because batches are executed on a platform that lacks a single point of failure. This can drastically reduce the number of process interruptions caused by hardware failures.
- Moving batch execution into the controller level can actually streamline the process itself, and eliminate the need for plants to purchase—not to mention install or maintain—additional batch hardware or software packages.

Using modern-controller-based batch control systems (e.g., Honeywell's Experion Batch Manage) allows the execution of batch and unit procedures directly in the controller to reduce system complexity, and improve availability and speed of execution. This can lower costs by 5 percent in improved productivity and 3 percent in reduced raw materials.

Chris Morse

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Cut the cable

My comments concern the otherwise well-written article by Lee Neitzel entitled "Six steps to control systems cybersecurity" in your May/June issue. His suggestions reflect the typical attitude by many, that when one firewall does not work, then add another and so on. Rather than use four firewalls and other expensive paraphernalia, I suggest you simply cut the cable to the Internet. This will give you the best and least-expensive security.

Hans D. Baumann

Silver money train

As you mentioned in "Talk to me" [July/August 2013 *InTech*], the industry is suffering from the ever-growing gap between skilled employees and positions in the petroleum industry. The answer to correcting this revolves around one single thing that could result in many different approaches—money. The industry professionals believe they are saving money by saying there are no skilled employees available. This is a scam to divert attention and blame, as they reap the benefits of raises and pats on the back. As a college student attending Lee College in Baytown, Texas, I can see clearly; please follow the pictures that I am going to paint for you.

First, the industry refuses to hire college graduates because of the lack of experience. Almost every position being advertised requires at least three-to-five years' experience.

Second, these so-called educated management professionals are not learning from the plant tragedies of history and all the lives lost. The mistakes were made by not having qualified, skilled employees and the "silver-bullet" layoffs and refusal to hire and train people to have a formally qualified understanding.

This is what I call the "magic silver bullet." Industry leaders can either spend the money to hire college graduates and train them correctly to keep a productive and more trouble-free plant that can be classified as "lean tuning," or they can increase the pay scale to attract qualified employee candidates for the open positions.

So, as you mentioned the silver-bullet syndrome, may I please add that you can only get a silver bullet by a single way, and that is what the goal of this article was truly about: money. Maybe you should have considered naming your article "The silver money train."

Troy Riddle



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Water and wastewater industry cyberattacks increasing



According to data in the Repository for Industrial Security Incidents (RISI) database, the transportation and water and wastewater industry sectors have both experienced large increases in reported cybersecurity incidents in recent years: 160 percent and 60 percent, respectively. These findings and many more were published in the *2013 Annual Report on Cyber Security Incidents and Trends Affecting Industrial Control Systems.*

RISI is an industry-wide repository for collecting, analyzing, and sharing information regarding cybersecurity incidents that directly affect industrial control and supervisory control and data acquisition (SCADA) systems. Industrial automation system suppliers, end users, and international government agencies and research institutes have used RISI data for business justification, risk analysis, and a variety of research projects.

Industrial control system and SCADA security has been a serious concern for more than a decade, but has come under increased scrutiny following the discovery of the Stuxnet virus in 2010, the Duqu virus in 2011, and the Shamoon virus in 2012. All of these viruses specifically targeted industrial control systems.

The 2013 report includes detailed analysis of the 240 incidents recorded in the RISI database from 2001 through 2012. The analysis identifies where and when the incidents occurred, the types of incidents, and the threat agents that executed them, including the methods and techniques used to gain entry. The financial and operational impacts on the "victims" are also analyzed.

The report includes detailed results and analysis from the second annual RISI Control System Security Benchmark Survey. The survey data provides very interesting insight into the current state of control system security, especially when compared with the data regarding actual incidents. For example, RISI data indicates that 33 percent of all industrial control system security incidents were perpetrated through remote access. This data is supported by 48 percent of survey respondents reporting that remote access to the controls systems is allowed at their facilities.

Twelve control system integrators join CSIA

The Control System Integrators Association (CSIA) announces the following control system integration companies have joined the association:

- Cogent Industrial Technologies, Richmond, British Columbia, Canada
- Fosdick & Hilmer Inc., Cincinnati, Ohio, U.S.
- Ingautin de Colombia Ltda, Bogota, D.C., Colombia
- Intecol SA, Medellin, Antioquia, Colombia
- Intelligent Systems Integrator EST, Riyadh, Saudi Arabia
- IS International Services LLC, Lawrenceville, Ga., U.S.
- Kubica Corp., Novi, Mich., U.S.
- Matador Controls LLC, Palm Bay, Fla., U.S.
- Outbound Technologies, New Hudson, Mich., U.S.
- Proctek Inc., Bakersfield, Calif., U.S.
- Superior Control Systems Inc., Tempe, Ariz., U.S.
- Trutegra, Charlotte, N.C., U.S.

CSIA helps members improve their business skills, provides a forum to share industry expertise, and promotes the benefits of hiring a certified control system integrator. CSIA has more than 400 members in 27 countries.

Improved maintenance with portable vibration analyzer

Kimberly Clark's Loudon, Tenn., paper mill was one of the first plants to test Emerson's CSI 2140 machinery health analyzer. After several weeks of using the new vibration analyzer, the plant reported faster completion of their normal maintenance routes, easier and quicker data analysis, and an overall more comfortable experience working with the unit in the field.

Multiple facilities like Kimberly Clark invested their time and experience in the beta test process for Emerson. Their real-world applications test the results users will achieve with the analyzer. "It's very quick, especially with slow-speed equipment," said Dennis Keaton, maintenance mechanic at Kimberly Clark's Loudon paper mill. "We run 1200 to 1300 points per week, so the speed of setting up and taking readings is very important."

"It's very quick, especially with slow-speed equipment."
—Keaton

An integrated analysis tool provides the complete picture of asset reliability. With the CSI 2140, vibration tools coexist with balancing, temperature, trends, and motor diagnostic tools. In addition, four-channel vibration analysis capabilities deliver insight into operating conditions to help correctly diagnose complex machinery problems. Embedded intelligence guided Kimberly Clark technicians through testing and analytics. Even less-experienced technicians could confidently monitor and diagnose equipment issues with minimal training.

In addition, the human-centered approach to the design of the CSI 2140 made an immediate impact. At Kimberly Clark, with approximately 8,400 measurements per month, the smaller size of the analyzer made routes more comfortable. Measurements in the dark areas of the mill were much easier with the brighter screen.

Automation by the Numbers

This content is courtesy of Automation.com

$7.88 billion

According to a radio-frequency identification (RFID) sector survey by IDTechEx Research, the RFID market will increase from $6.98 billion in 2012 to **$7.88 billion** in 2013. It will reach $23.4 billion in 2020. The market includes tags, readers, and software and services for RFID cards, labels, fobs, and all other form factors for both passive and active RFID.

The market for RFID has grown steadily despite the economic meltdown due to the diverse nature of its applications, from tagging retail apparel to transport ticketing to animals. Governments drive most RFID orders as they improve efficiency (transit systems), safety (passport tagging), and protective industries (animal tagging). Since 2000 there has been a



strong push to use passive RFID to improve supply chain visibility, with investment in new RFID technologies, new standards, and much publicity. As with most new technologies, initially some aspects were overly hyped and demand was not in sync with capacity, but at the beginning of 2010 the industry emerged from the hype cycle and entered a period of rapid growth and profitability for some. There are different rates of growth for different applications and many challenges—but opportunities still exist. In total, IDTechEx finds that 5.9 billion tags will be sold in 2013, versus 4.8 billion in 2012.

350

At the Audi factory in Gyor, Hungary, Siemens constructed a fully automated high-bay warehouse that acts as a central body buffer and occupies a key position within the production logistics of the plant. Siemens developed the controller of the storage and retrieval machines and the conveyor system for the body buffer on the basis of its Simatic S7 controllers. The bodies are conveyed by means of **350** Siemens electric motors together with frequency converters. The integrated-drive system has energy-saving motors and can recover and store kinetic energy. This means that it captures braking energy, stores it, and makes it available to other drives. The result: fewer undesirable network perturbations and lower power consumption. In addition, Siemens set up an RFID system that uniquely identifies each body stored in the body buffer and on its way through production.

54

The PROFI Interface Center (PIC) has completed a "wall of fame" in its test lab that features PROFINET devices it has tested and certified. The PIC is part of the global network of competence centers, training centers, and test labs established by PI, the umbrella organization responsible for PROFIBUS and PROFINET internationally. Certification of PROFINET devices is required through PI. This ensures interoperability between products, and the labs can support device vendors in case of conformance questions.

Manager of the PIC, Torsten Paulsen, commented, "We created this installation to feature all of the devices we have certified over the years and put them to use as a 'real world' test bed and reference network. We currently have **54** nodes and add more constantly." A wide range of devices are incorporated onto the wall including remote input/output, drives, vision systems, human-machine interfaces, wireless, embedded diagnostics, infrastructure, and security devices. To date, more than 500 different PROFINET-enabled products have been tested and certified worldwide.

$40 million

Endress+Hauser is set to serve its customers in North and South America with more U.S.-made products. The Swiss specialist for measurement engineering and process automation has invested more than **$40 million** in expanding its plants for flow, level, and pressure measurement engineering in Greenwood, Ind. Producing, assembling, and calibrating at sites all over the world is one of Endress+Hauser's central business strategies. One important factor for this concept to work is globally uniform quality standards. The systems for calibrating flowmeters now include the latest technology for gas calibration.

$2.765 million

Representative David Price (D-N.C.), the ranking member of the House Homeland Security Appropriations subcommittee, announced that ABB Inc. has been awarded a **$2.765 million** Department of Energy (DOE) grant to develop technology to help the energy sector enhance its cybersecurity. The project is one of 11 recently announced DOE grants totaling more than $30 million. ABB will lead a project to develop a system that allows electrical substation devices to work together to validate the integrity of communications along a power distribution network.

The grant awards are a part of the Obama administration and Congress' commitment to protecting the nation's critical infrastructure from cybersecurity threats.



Next frontier:
Operator-automation relationship



Preclude operator error, improve safety and profitability

By Eddie Habibi

Industrial control systems are complex designs, but they do not give much consideration to the needs of the human operator. This oversight has caused serious and costly accidents. By empowering operators with the right human-automation relationship tools, companies will achieve and exceed their goals.

Unprecedented prosperity

The world population is more prosperous today than ever before. The average global per capita gross domestic product has grown by nearly tenfold during the past 140 years. By comparison, this measure was practically flat from 1600 to 1875, the period before the Industrial Revolution. The fastest growth in productivity emerged after World War II and accelerated during the past three decades. The U.S. economy exploded, growing from $2.8 trillion in 1980 to more than $15.5 trillion in 2012. Human ingenuity and innovation made possible by connectivity, information sharing, and collaboration form the basis of this relative prosperity.

Unintended consequences

Specific to industrial production, much of the productivity growth in recent decades is attributed to advances in process automation technologies such as the distributed control system (DCS), the historian, model-based controls, and production optimization. Without much incremental investment in plant equipment, automation technologies have substantially reduced variability and cost, improved throughput and quality, extended asset reliability, and delivered incredible financial returns on investment.

Industrial control systems are complex. They process and generate significant information in real time, and were designed without much thought for the needs of the human operator. Serious and costly accidents have been the consequences of this oversight. Chernobyl, Bhopal, Three Mile Island, Piper Alpha, and Texas City were some of the worst industrial accidents in recent history. Human error is cited as the root cause or a major contributing factor in every one.

Do not blame the operator

Unfortunately, there is a tendency to automatically blame the human operator. Often, the operator is unjustly blamed for such accidents. An operator inherits a production facility as it was designed and constructed. Traditionally, the automation system configuration and the

resulting alarms and interfaces are designed and implemented by engineers unfamiliar with human-factors design. It is not unusual for an operator to receive a barrage of alarms during a process upset, or to search through five displays before finding the relevant information related to the situation. Operators, like most everyone else, want to succeed at their jobs.

Understanding human error

Human error may simply be described as the failure to carry out a given task (or the performance of an undesired action) that could result in disruption of scheduled operations or damage to equipment and property. In process operations, human error occurs when an operator fails to take proper action (the right action at the right time) or takes no action when action is required.

Human error can be divided into two general categories:

Intentional errors occur when an operator deliberately performs, or chooses not to perform, a task. In almost every case, intentional errors are *not malicious*. Typically, when committing intentional errors, operators believe their actions are correct or more appropriate than what the standard operating procedure calls for. For example, an intentional error occurs when an operator deviates from a written procedure, falsely believing that the procedure is incorrect.

Unintentional errors occur when a worker unwittingly performs, or chooses not to perform, a task. For example, an operator might unintentionally enter 47 percent instead of 4.7 percent when moving a control valve. Another example of an unintentional error is when an operator does not detect the emergence of an abnormal situation and fails to take action entirely. These human errors are generally referred to as *accidents*.

Two factors influence an operator's decision-making process:

Internal factors that reside within the operator, such as tacit knowledge, experience, cognitive abilities, fatigue, and work ethic, affect his or her ability to process information and act promptly. Management can influence the internal factors through rigorous hiring practices, training, policies and procedures, and overall organizational culture.

External factors are outside influences that act upon the operator and impact his or her decision process. They include the physical environment and the information flow. The control room ergonomics such as lighting, temperature, and noise are components of the physical environment. Sources of information flow encompass the telephone, the radio, conversations, and the automation and information systems.

As an industry, we have to believe that *all human error is preventable*. Otherwise, the concept of zero accidents, which is the vision of every industrial company, is an unattainable goal.

FAST FORWARD

- Industrial control systems have been designed without much consideration for the needs of the human operator, resulting in serious and costly accidents.
- Operators, like most everyone else, want to succeed at their jobs, but poor human engineering results in confusing displays and alarms.
- Operators are true knowledge workers, and they can be empowered by applying the right human-automation relationship tools.

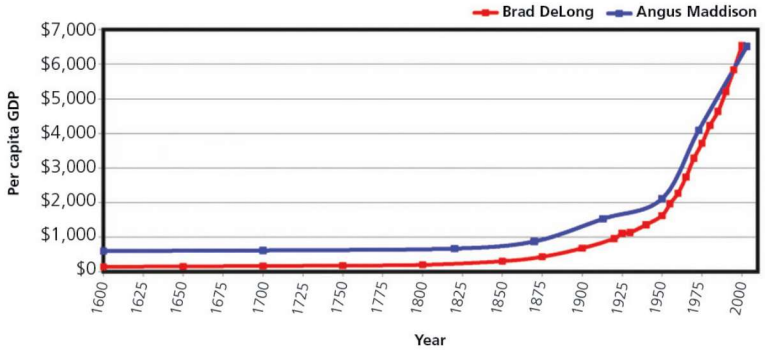
The knowledge-worker operator

The console operator is the final human element to interact with the production process in real time. The decisions and actions of the operator directly affect process safety and company profitability. Thus, the operator is the most critical element in the success of any industrial company. When the operator has a bad day, everyone in the company has a bad day, from the chief executive officer to the last shareholder. More importantly, if the bad day involves casualties or a major environmental excursion, the surrounding community suffers too. The performance of no other individual has such a direct impact on the profitability and the reputation of a company.

It is time for the industry to recognize the role of the operator as mission critical and deserving of proper respect and investment. Peter F. Drucker, the father of modern management theory, described a knowledge worker as one who uses or creates information in the process of performing a job. Operators have to monitor, analyze, and take action based on thousands of data points in

World per capita GDP

1600–2003 (1990 International dollars)



Source: J. Bradford DeLong, "Estimating World GDP, One Million B.C. - Present" http://www.j-bradford-delong.net/TCEH/1998_Draft/World_Estimating_World_GDP.html. Accessed Mar 5, 2008; Angus maddison. "Contours of the World Economy, 1 - 2030 AD: Essays in Macro-Economic History." New York: Oxford University Press, 2007.382. © 2008, Matthew W. Kruse

complex, fast-moving, real-time situations. Operators are the ultimate automation industry knowledge workers.

Companies such as BASF place significant value on the role and the contribution of their operators. "We recognize the important role of our knowledge worker operators in the success of our company," says Chris Witte, senior vice president and site manager at BASF Freeport. "We are investing in technologies such as alarm management, high performance HMI, and automation asset management to help improve safety and asset reliability."



The historical marker in front of the Three Mile Island nuclear power plant reminds us of the role of human error in that accident.

Situation awareness: the operator's blink

Situation awareness refers to the operator's comprehension of the condition of a process unit at any moment based on his or her personal knowledge, cognitive abilities, and the information presented to him or her. Proper situation awareness is critical to successful decision making in complex and dynamic professions such as aviation, firefighting, and industrial operations. Operators with proper situation awareness are more successful at analyzing information, identifying critical conditions, and taking proper actions to mitigate undesired consequences.

In his groundbreaking book, *Blink: The Power of Thinking Without Thinking*, Malcolm Gladwell tells the story of a firefighter in Cleveland, Ohio, and how he and his team escape a deadly fire trap. After numerous failed attempts to extinguish the fire in the kitchen of a one-story apartment in Cleveland, the lieutenant ordered his men out of the building. As soon as they exited, the kitchen floor where the men had been standing collapsed. Had the men not left the building when they did, they would have been swallowed by the fire. At first, the lieutenant refers to his timely judgment as "gut feel" or "ESP," but after an extensive interview it is revealed that the experience and the cognitive abilities of the firefighter helped him analyze the situation and arrive at the correct decision in a split second. The lieutenant had realized that "something was wrong" based on three distinct observations: the fire was very hot, it was stubborn and would not go out, and there was an uncharacteristic silence in the kitchen for the intensity of the fire. That split-second decision is what Gladwell calls "blink." Blink is that moment of absolute clarity, when the firefighter thin sliced the information presented to him and, based on his experience, made the correct call.

Experienced operators, similarly to Gladwell's firefighter, have the ability to absorb large amounts of information, filter it, and connect the dots, and then make split-second decisions and respond to seemingly impossible situations. Proper situation awareness in the control room leads to improved operator comprehension of complex situations and fast and accurate decision making.

During abnormal situations, a typical control system may present hundreds or thousands of data points to the operator. This is indicative of a bad relationship between the operator and the automation system. Management must take steps to optimize the operator-automation relationship.

The human-automation relationship

The most differentiating competitive advantage in the consumer electronics market is the *user experience*. Apple significantly raised the bar with its design of the iPhone. In the past seven years, the Apple iPhone has gone from being the new kid on the block to selling more units in one weekend than BlackBerry did in the prior three months. The user experience, more than any other factor, made the iPhone the darling of the mobile device market.

In contrast, we can confidently claim that, at least until recently, user experience and interface design have not been viewed by automation vendors as a competitive advantage. The primary focus, instead, has been on making faster and better process controllers. Consequently, during a process disturbance when human intervention is required to rescue the plant, most control systems generate an inordinate number of alarms, creating sensory overload and making the situation worse for the operator.

The power and process industries today face a serious challenge that is the direct consequence of failures in the *operator-automation relationship* design.

Challenges in the operator-automation relationship arise from incongruity between the automation system's user interface and the operator's ability to process information and take action toward a desired outcome.

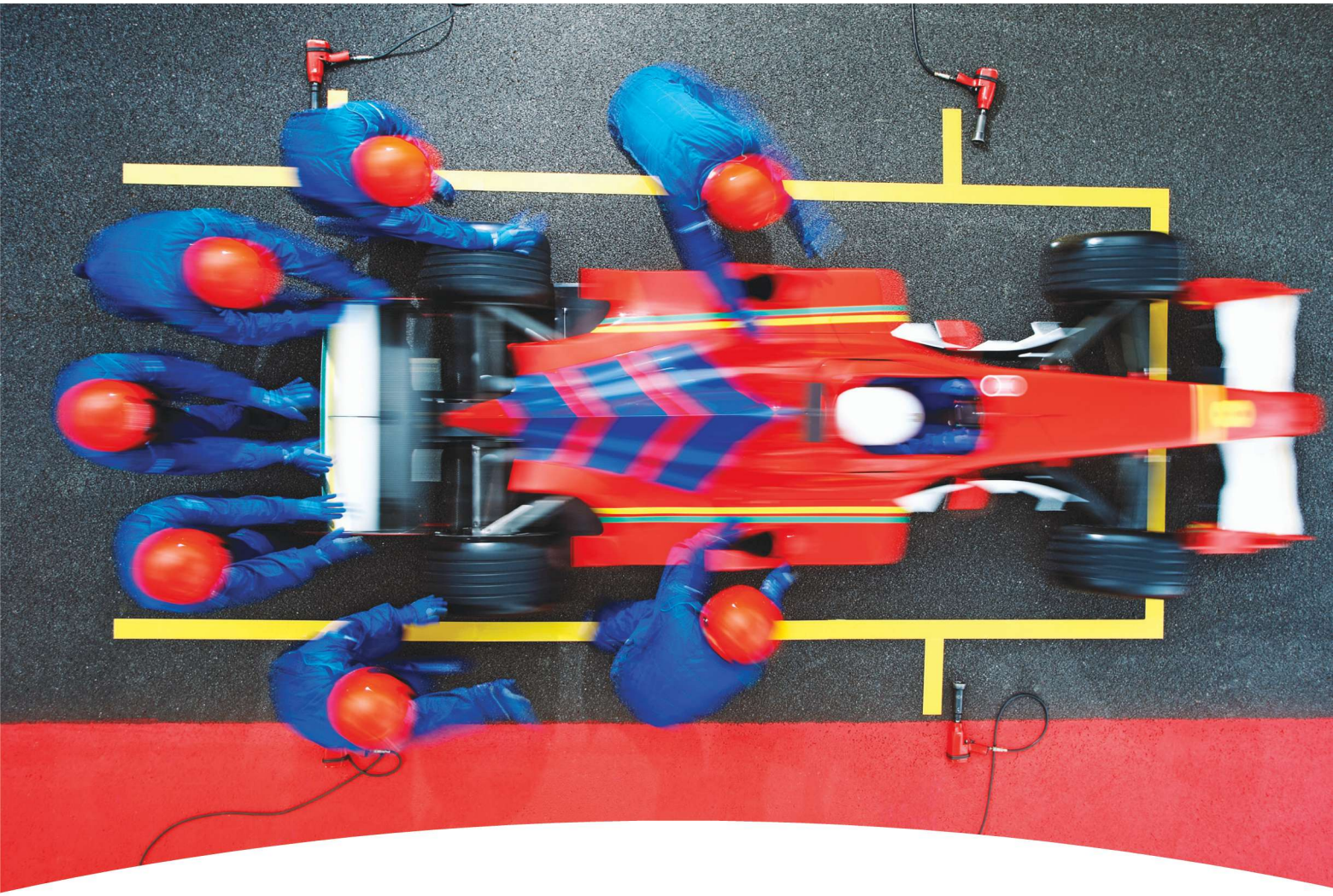
Improving the human-automation relationship for the console operator begins with the user interface. The operator interface is also the proper portal for a comprehensive and fully integrated decision-support system that enables knowledge retention and collaboration in real time.

A number of important elements must come together to create an effective human-automation relationship environment. These include:

High-performance HMI

The console graphical interface is the operator's window to the process. One of the major user-interface challenges today is the absence of an effective "big picture" overview display. Unlike the wall-mounted instrument panel of the past that provided at-a-glance situation awareness of plant conditions, con-

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trol system consoles today provide a "key-hole" view of the process through 60-to-100 individual displays. Most operator displays are cluttered, lack hierarchy, provide no pattern recognition, and use colors indiscriminately.

An effective resolution to the deficiencies of today's operator displays is high-performance human-machine interface (HP HMI). HP HMI displays follow essential human-factor design principles that include, among other related display design best practices:

- Three levels of display hierarchy
- Grey-scale colors
- Chunking and grouping of information
- Pattern recognition objects
- Simple and intuitive navigation, including pan and zoom

HP-HMI-based displays are minimalist in detail and rich in useful information. True HP HMI displays also include the capability to easily integrate critically useful information like alarm response documentation, control logic interlocks, checklists, and operating procedures.

Alarm management

The purpose of an alarm is to inform the operator that action must be taken to mitigate an undesired situation. An effective alarm must be unambiguous, unique, timely, actionable, and with proper priority to convey the correct level of urgency to the operator. Results of alarm system performance studies conducted by PAS indicate that most alarm systems perform poorly under normal conditions and, even worse, become a hindrance to the operator during process upsets. Alarm floods, disabled alarms, and long-standing stale alarms are three of the top culprits that create confusion in the control room and cause operator error.

The issue of ineffective alarm systems has been around since the advent of the DCS. However, in recent years, alarm management optimization has become an industry best practice and an opportunity to improve plant safety and profitability.

A robust alarm management strategy must include software to automatically capture, archive, analyze, and report the performance of the alarm system. An integrated documentation and rationalization engine is needed to facilitate proper engineering of priorities and trip settings and to capture causes, consequences, and corrective actions for each alarm. A master alarm database with audit and enforcement capability ensures the integrity of the reengineered alarm system. Dynamic state-based alarming is an essential part of an integrated alarm system that automatically changes the alarm settings to properly match the state of the process.

Boundary management

Capturing, monitoring, and performing strict change control on limits of operability are crucial to safe operations. Typically, process and equipment boundaries such as alarms, the safety system trip point, pressure relief specifications, and other limits are maintained by different organizations within a plant. Operators usually have visibility to alarm limits only. There are two issues with the way boundary information is managed in today's operations: it is nearly impossible to ensure the integrity of the information for all the boundaries associated with a given piece of equipment; and operators rarely have access to all the related boundary information in real time.

A consolidated approach is required to manage the integrity of operational and safety boundaries; this includes strict management of change and periodic audits. Furthermore, integrated operator interfaces are needed to give the operator an at-a-glance view of all the boundary limits associated with a piece of equipment.

Procedural automation

Standard operating procedures (SOPs) are step-by-step written instructions that guide operators in the uniform operation of a process. Many industrial incidents happen when plant personnel either deviate from or ignore the written SOP. Other times, incidents occur when personnel follow a procedure that is out-of-date.

Procedural automation is an effective method for mitigating most incidents that are caused by circumventing established SOPs. In 2010, ISA launched a new standards committee on procedural automation for continuous processes. ISA-106 reflects the combined best practices of several global manufacturing companies such as Dow Chemical Company, Aramco, Chevron, and ConocoPhillips.

"For over thirty years, Dow Chemical has used a proprietary control system [MOD] to implement state-based controls methodology to improve operator performance, enforce our operating discipline, and significantly enhance process safety throughout our company," said Yahya Nazer, Ph.D., manufacturing and engineering consultant at Dow Chemical Company. "We see ISA-106 standards for procedural automation as an avenue for sharing our experience with other companies."

Many automation system suppliers are integrating automated procedures into their new platform designs.

Integrated information portal

An integrated information portal provides simple access to supplemental information such as operating procedures, checklists, and instructional videos that improve the operator's situation awareness. It is also a platform for knowledge retention and collaboration.

Below is a summary of seven key elements of an effective knowledge retention and collaboration platform that enables the integrated information portal within the operator's work environment.

1. Aggregate: capture existing information or *explicit knowledge* that is available in a digital form (e.g., design drawings, procedures, training videos, and control system logic)
2. Author: allow operators and engineers to enter contextual information and expand the knowledge base
3. Contextualize: automatically recognize relationships and give context to information from disparate sources
4. Tag: allow users to categorize or classify related points of information by adding user-defined tags to multiple information objects
5. Search: allow Google-like searches on the process control and plant information networks. It is imperative

that search results be concise, accurate, and previously vetted by designated subject matter experts.

6. Alert: allow operators to subscribe to and receive alerts that would notify them of an event, a process condition, or a task
7. Recommend: similar to the Amazon books' recommendation "people who bought this book also looked at this other," the integrated information portal must provide useful recommendations to help the operator improve his or her decision process. For example, the system should recognize when an operator selects a pump to start. It should then automatically present to the operator a checklist to follow.

Death of the traditional logbook

The traditional stand-alone operator logbook, whether in paper or electronic form, is obsolete. The information communicated through the operator logbook is highly valuable and interdependent with other production systems. The operator logbook can no longer be viewed as a stand-alone system. In fact, the functionality of an operator logbook must be included within the integrated information portal.

Summary

Operators are true knowledge workers whose performance directly affects the company's profitability and reputation. It is the responsibility of the power and process industries to recognize and empower the operator in the same way that the airline industry and the medical community recognize and empower pilots and emergency room surgeons, respectively. By empowering the operator with the right human-automation relationship tools, the company will have a solid foundation for safe production.

Fortunately, the new generation of industry executives—those who grew through the ranks during the digital revolution—understands the critical role of the operator in the success of their organizations. They keenly recognize that the role of the operator is less about turning valves and making set point adjustments and more about assessing situations and making economic decisions in real time, often under adverse conditions. They are investing in their operators and in new enabling technologies to optimize the performance of their organizations.

ABOUT THE AUTHOR

Eddie Habibi (eddie@pas.com) is the founder and CEO of Houston-based PAS, a supplier of human reliability software and services to the power and processing industries worldwide. He is a thought leader in the areas of operator effectiveness, automation and information integrity, and web-enabled knowledge retention and collaboration technologies. Habibi has an engineering degree from the University of Houston and an MBA from the University of St. Thomas. He is the coauthor of two popular books on industrial operator effectiveness: *The Alarm Management Handbook* and *The High Performance HMI Handbook*.

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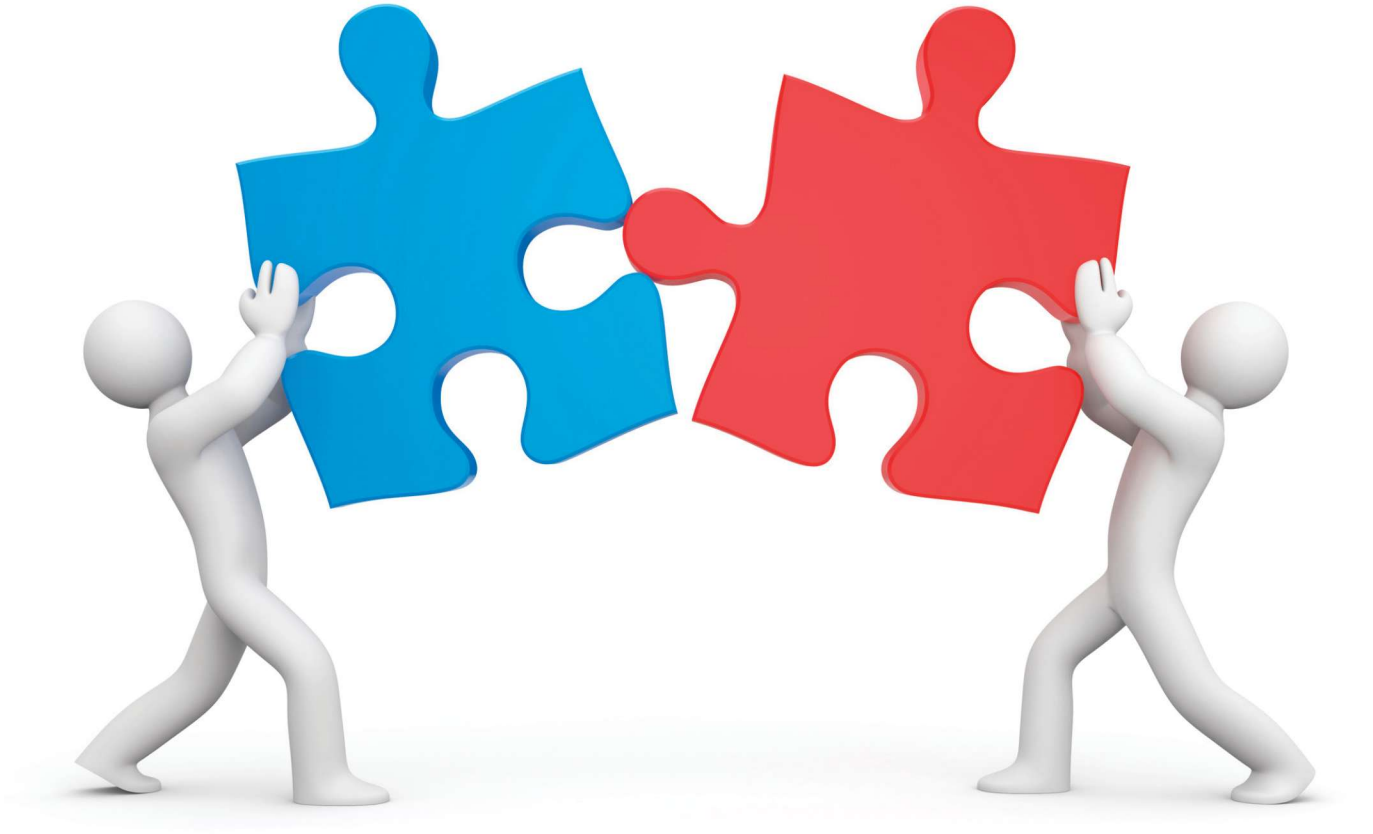
Related ISA Standards Committees:

ISA18 – Alarms; ISA101 – HMI; ISA106 – Procedure Automation
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Big changes ahead for Fieldbus Foundation

on the eve of twentieth anniversary



Fieldbus Foundation to focus on usability, cross-industry cooperation, and the right work processes

By Larry O'Brien

The Fieldbus Foundation will celebrate its twentieth year in 2014. It is also a year of transformation into what could very well be a new organization entirely, with some very exciting developments for the world of intelligent devices and plant asset management. Today, the Fieldbus Foundation has more than 500 members and affiliates around the world. An installed base of about $10 billion in products and systems controls critical processes in the world's largest plants.

Because FOUNDATION fieldbus is a truly open standard (anybody can purchase our specification or become a member), we can continue to adapt to new technologies as they come along and evolve our specification to improve what we already have. Some of the key ways we are moving forward for 2014 and beyond include discussions with the HART Communication Foundation regarding a potential merger of the two organizations, the development and release of the Field Device Integration (FDI) specification, a continuous improvement initiative called Project Gemstone focused on enhanced usability, continued work on the ISA108 standard for work processes that use diagnostic data from intelligent field devices, and the further development of the FOUNDATION for Remote Operations Management (ROM) specification.

FAST FORWARD

- Potential merger between Fieldbus Foundation and HART Communication Foundation creates industry buzz.
- Fieldbus Foundation focuses on enhancing usability for developers and end users.
- ISA108 committee works to give users the necessary work processes to use digital diagnostics—the right way.

Combining efforts

In our recent press release, the Fieldbus Foundation and HART Communication Foundation have signed a memorandum of understanding to look into the possibility of forming a new foundation focused on the requirements of digital communications and intelligent field devices for the process industries. This seems to be a natural fit for both organizations, and indeed many people in the industry have spoken to the Fieldbus Foundation over the past several years about the benefits of combining both organizations, which have a long history of cooperation. Combined efforts include the electronic device descriptive language cooperation team and, most recently, FDI.

A combined organization of Fieldbus Foundation and HART technologies could better leverage the complementary benefits of the technologies. The new organization would create more cooperation and collaboration. In addition, we would have improved economies of scale through merging training and education; seminars; testing and registration; participation at trade shows, conferences, and events; online presence; and social media strategies. While these efforts are still in the exploratory phase and there is still a lot of work to do, this is obviously the biggest and most important development we are focusing on for 2014, and we will keep the process automation community abreast of our progress.

In the meantime, however, we continue to pursue a number of other initiatives, all of which are designed to make things easier for both end users and developers in terms of device development, usability, and getting the maximum benefit from FOUNDATION technology. These include the FDI specification; Project Gemstone, which is focused on usability; our involvement with the ISA108 standard for work processes related to intelligent device management; our field demonstration program for FOUNDATION for ROM technology; and our impending testing and registration of devices and hosts that are compliant with our FOUNDATION for safety-instrumented functions technology.

Field Device Integration

At the core of the FDI specification is the FDI device package, which includes everything a host system needs to integrate an intelligent device. With FDI, each device is represented by a single package that can scale according to the complexities and requirements of each device. Each FDI device package contains a mandatory device description that provides parameter definitions, structure for the parameters for context-specific views, and automated work processes for device procedures such as calibration. FDI device packages may also include user interface plug-ins, software components that support advanced device setup, and diagnostic functions. Product manuals, documentation, images, electronic certifications, and other attachments may also be delivered in the FDI device package.

FDI device packages make it easier for automation suppliers to develop and integrate intelligent devices, because suppliers only need to create a single, unified package for each intelligent device that can work with all host systems and tools. This reduces overall development costs, while preserving and expanding existing functionality. Users will also find it easier to manage information from intelligent devices with a single device package, instead of juggling different technologies and investing significant capital in custom integration efforts to connect multiple technology platforms.

Project Gemstone

Project Gemstone is a continuous improvement project that the Fieldbus Foundation launched this past year. It focuses on making FOUNDATION fieldbus easier to use for both end users and suppliers. FDI is a major component of Project Gemstone, but perhaps more important is the formation of a new group focused exclusively on usability. Like many of the activities at Field-

bus Foundation, Project Gemstone is greatly influenced by the requirements of our End User Advisory Council. End users have probably noticed the proliferation of such initiatives at many supplier companies, such as Emerson Process Management's "human-centered design" initiatives, and our usability efforts are very similar. The whole idea is to allow people to focus more on how FOUNDATION technology can help them in their business versus managing the technology itself. The ultimate goal is making FOUNDATION technology easier to use than 4–20 mA analog technology.

A great example of this is making device replacement easier. We have already made enhancements to our interoperability test kit that makes it easier for end users to replace "like-for-like" devices. In other words, if you are replacing a device with a newer revision level of the same make and model, we have automated this procedure to restore the configuration of an existing field device in a newer version of that instrument without manual intervention. Our next step is to tackle the issue of "unlike" device replacement, where the end user is replacing a device with the same type of device, but from a different manufacturer.

FF standard diagnostic alarm	NE107 status signal (5.1)
SD_FAIL_ALM	Failure
SD_OFFSPEC_ALM	Out of specification
SD_MAINT_ALM	Maintenance request
SD_CHECK_ALM	Function check

One of the primary reasons end users do not realize the benefits of intelligent devices is because they have not changed their existing maintenance work processes from the same ones they used with the old 4–20 mA devices. Most users do not even know what work processes should be adopted, or how those work processes should flow. The new devices and applications are installed, but the operators and technicians stick to their old way of doing things, their old preventative or routine maintenance practices. They never really take advantage of the huge amount of information available to them.

Ask most end users why they made the investment in FOUNDATION technology, and they will tell you it was because of the advanced diagnostics capabilities of fieldbus devices.

ISA108 puts work processes in user's hands

Ask most end users why they made the investment in FOUNDATION technology, and they will tell you it was because of the advanced diagnostics capabilities of fieldbus devices. According to a recent survey by the Fieldbus Foundation and ARC Advisory Group, however, many users still do not invest in plant asset management applications that can truly allow them to take advantage of the diagnostic information from their intelligent devices and create a condition-based or predictive asset management strategy.

The ISA108 standards committee was formed in 2012 to address this issue and provide users with the knowledge they need to implement the right work processes and really get the life-cycle benefit out of their intelligent devices. Let us be clear that ISA108 is not a FOUNDATION fieldbus-centric committee, although the Foundation is heavily involved. This project applies to anyone who is using intelligent field devices on any network. If you are interested in participating in the ISA108 committee, please contact Charley Robinson of ISA Standards (crobinson@isa.org).

Making sense of the diagnostics storm

Better diagnostics is one of the great promises of digital networking technology for field devices. All "intelligent" devices can provide some level of diagnostics. There are several key differences with FOUNDATION fieldbus that make its ability to provide diagnostics unique. Two of these differences are the sheer volume and frequency of diagnostic data provided. FOUNDATION devices can handle multivariable measurements and transmission of multiple diagnostic data at the same time.

The diagnostics do not stop at the sensor or actuator. Diagnostic data can be provided for electronics failures, configuration or servicing failures that are primarily human intervention issues, and application issues or process issues that affect the measurement. These multiple levels of diagnostics add up, until you could reach over 20 diagnostic parameters for FOUNDATION fieldbus devices, with more complex devices and actuators having hundreds of parameters. To get useful information out of these diagnostics, you must have a good way to manage and organize this information. Some diagnostic parameters are manufacturer-specific, and at the end of the day, this can be a real challenge. Parameters from device to device may not be the same. Based on the root cause of a diagnostic, however, diagnostics can be categorized or assigned to different functional areas, such as electronics, configuration, or application.

The FOUNDATION Fieldbus Field Diagnostics Profile specification was defined to make it easier for end users to access and configure the diagnostics in fieldbus devices, regardless of the manufacturer's device or system. The diagnostic profile includes a standard and open interface for reporting all device alarm conditions and has a way to categorize alert conditions by severity. The technology facilitates routing alerts to appropriate consoles based on severity categories selected by the end user. In other words, it sends the right information to the right person at the right time without flooding the operator with alarms that are irrelevant to his or her duties. The field diagnostic profile also

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gives recommended corrective actions and detailed help, as well as an indication of the overall health of the device. The Field Diagnostic Profile specification puts all the necessary mechanisms in place to provide context to diagnostic data and turn it into useful information.

A major part of the diagnostics specification is the inclusion of the NAMUR NE107 recommendations for providing diagnostic information in context to the right people at the right time. NAMUR NE107 categorizes internal diagnostics into four standard status signals: failure, function check, out of specification, and maintenance required (also known as FCSM). Each category can also contain greater detail. In the case of failure, for example, can the failure be traced to the device or the process? Is maintenance required immediately, or is the requirement for long-term maintenance?

NE107 standard status signals

The ultimate result is a series of new field diagnostic alarms that correspond to the four primary diagnostic categories outlined by NAMUR NE107. Several standardized and therefore manufacturer-independent parameters are available to configure the NAMUR category, the priorities, and the filter mechanisms for the alarms. With NAMUR NE107 diagnostics built in, you can turn off diagnostics you do not need or configure how the diagnostics are reported. This supports the configurability mandate of NE107. Providing recommended actions and enabling simulation allows the information to be presented in greater context.

FOUNDATION for ROM

Remote operations management is the management of automation assets and resources that are geographically dispersed. It is one of the fastest growing segments of the process automation business. Today, the ROM segment is plagued with a high degree of customization, solutions that are not easily configurable, and a break-and-fix mentality when it comes to asset management. FOUNDATION Fieldbus has now expanded into the rapidly growing world of ROM, from the digital oil field to pipelines, tank farms, and floating production storage and offloading facilities.

Several years ago, the Fieldbus Foundation began a new project that would extend the functionality and infrastructure of FOUNDATION fieldbus out to remote applications through the integration of remote input/output (I/O) and wired HART communications. We then expanded that project to include leading industrial wireless networks, such as ISA100.11a and WirelessHART. Working cooperatively with ISA, the Fieldbus Foundation has also specified a long-distance backhaul network technology based on standard high-speed Ethernet that connects remote assets with their command centers. Today, our overall FOUNDATION for ROM specification is nearly complete and ready to extend our capabilities to manage data to a limitless range of networks and devices in some of the world's most unforgiving applications. FOUNDATION for ROM is ready to address the rapidly changing business challenges of end users in remote operations.

The FOUNDATION for ROM specification can be embedded into a range of products, including remote terminal units (RTUs), controllers, and remote I/O modules.

The FOUNDATION for ROM specification can be embedded into a range of products, including remote terminal units (RTUs), controllers, and remote I/O modules. Once the FOUNDATION ROM specification is embedded in a device, however, its functionality expands to reflect a combination of the traditional functions found in wireless gateways, process controllers, and RTUs. FOUNDATION for ROM devices provide the protocol translation functions of a gateway, but they go beyond the traditional functions of a gateway. They have the ability to represent these devices as transducer blocks in the FOUNDATION fieldbus infrastructure and all the capabilities that it holds, including data management, alarms and events, data quality, and function block structure.

The Fieldbus Foundation, in partnership with our demonstration program sponsors, has been hard at work on a series of live field demonstrations to show the value of the technology. We had our first successful live field demo in April at the Petrobras Cenpes research facility in Rio de Janeiro, Brazil. Running on one of the many Cenpes pilot plants, the Petrobras field demonstration offered a look at the full functionality of FOUNDATION for ROM, including wireless device integration, remote I/O integration, and wireless backhaul capabilities. This demonstration was the first step in showing the capabilities of FOUNDATION for ROM before it is specified for commercial projects. We also performed a demo of the technology at the Automation 2013 trade show in Mumbai, India, in October, where we won the technology advancement award for the show. The equipment used on this demo will also be used at our next live field demonstration at the Reliance Jamnagar refinery, which is the largest single site refinery in the world and the largest FOUNDATION fieldbus installation in the world. This demo is scheduled to take place in the first half of 2014.

ABOUT THE AUTHOR

Larry O'Brien (larry.obrien@fieldbus.org) joined the Fieldbus Foundation as global marketing manager in April 2011. Before his job at the Foundation, O'Brien was research director for process automation at ARC Advisory Group, where he began work in 1993. As an industry analyst and market researcher, he covered the topics of process fieldbus, distributed control systems, process safety, the automation services business, and intelligent field instruments. O'Brien has authored or coauthored numerous market forecast reports, strategic-level advisory reports, and white papers for ARC and its clients, including all the major process automation suppliers. He has a B.A. from the University of Massachusetts at Lowell.

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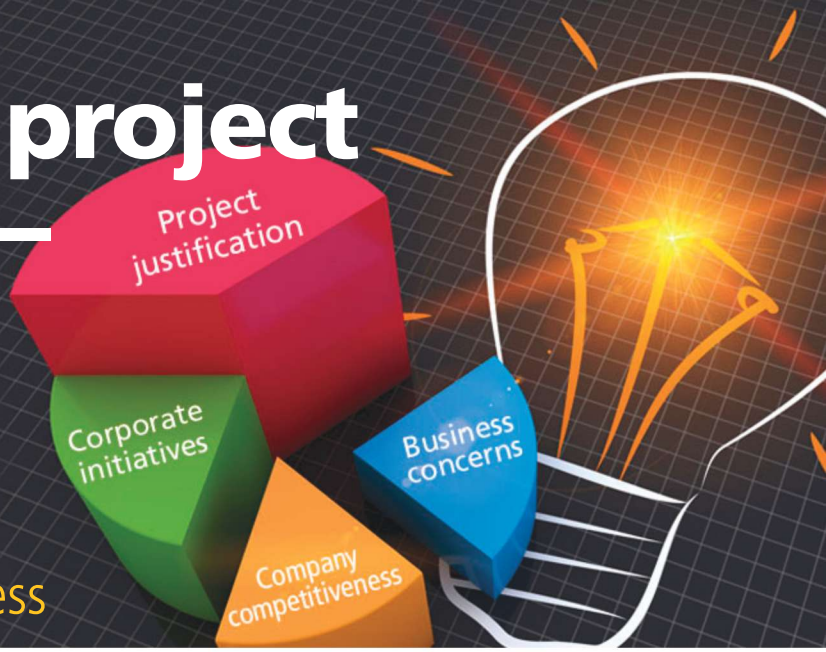
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Automation project justification – just about the math?

Project justification is critical to driving corporate competitiveness



By Dick Hill

Clearly, any new automation project requires economic justification. Although this often involves hard "dollars and cents" quantification of cost savings from increased production, reduced raw material costs, or increased uptime, it is also about "soft" benefits, such as reducing risks and improving agility. All contribute to making your company more competitive. This article will focus on the latter, while also highlighting considerations for the former. It will also share some best practices that ARC Advisory Group has identified.

When considering a change of any kind and researching the alternatives, it is very important to understand that your business is likely to be flooded with opportunities to invest capital and that change needs to be justified for the benefit of the company. Personal or department goals need to be subordinated to corporate goals. In the automation domain, a number of factors should be evaluated for their economic contributions to a justification. But it does not end there. Justification must take into account the broader opportunity to better position your company for future competition, likely involving greater agility, additional products, or improved quality.

Example items to consider in a justification

Maintenance and service costs: These costs will increase for older equipment. Spare parts may be difficult to find. Direct replacements for a component disappear. Those familiar with the technologies may become a rare commodity. In addition to increased costs of repair, older systems are likely to have a longer mean time to repair. If repair requires production shutdown, this downtime will clearly affect production throughput, and it will be important to know your cost of downtime.

Energy costs: Older systems typically use more energy. For example, fixed-speed motors are giving way to variable-speed drives for energy savings. Loose process control may cause cycling that can waste energy or affect product quality. Process control designs can be improved while also improving energy efficiency.

Recycling and sustainability: These factors may also be relevant in corporate culture or product costing. Recycling costs or construction materials may figure into the value of a system or its replacement.

Improved control dynamics: These will also lead to shorter start-ups, improved product quality, quicker product changeovers, and quicker shutdowns. These all have associated cost savings.

Removal of physical constraints: Modern control systems, for example, may improve space utilization even with expanded operations.

Improved data analytics: Today's advanced analytics can help streamline operations through system modeling. Perhaps a new solution can rely on data analytics—process modeling to remove the need for physical product sampling. An example of this is predictive emissions monitoring systems replacing continuous emissions monitoring systems in some applications.

Added flexibility: Existing systems may have been designed and implemented for a specific product and purpose, but are difficult to change and adapt to new product or business requirements. These requirements can include product variations or changes to batch or lot sizes. A new system can provide a great deal of flexibility, which could be a significant benefit for your business.

Now, for a couple of items that may not be as intuitive:

Upgrades to complex systems, such as a control system, often allow companies to make continuous improvements over time, plan for graceful degradation of an automation system, and perform future expansions in a well-managed,

modular fashion. For example, redundancy is typically a requirement in medium-to-large automation applications. But for smaller applications, the loss of a display console may be compensated for by a backup Web-based human-machine interface direct to a controller. This level of backup can come at a drastically reduced price, while also delivering added flexibility and functionality on a day-to-day basis.

A concentration on modularity for a new system can deliver a great many new features that can be surprisingly easy to implement. Perhaps you will want a text message while you troubleshoot a challenging issue. Perhaps you will want a special report while you are making some control enhancements to generate "before-and-after" analytics. Or, perhaps you will want to plan ahead for integrating additional sensors that you know will be coming with a new enhancement down the line. Relying on standards and leaving the door open to third-party products and interfaces and ad-hoc and short-term enhancements outside the core functionality of a system could help your company adapt to changing needs. Selecting the right technologies and data infrastructures and focusing on integrating best-in-class components can address your current needs well, while also significantly improving your ability to respond to future and transient requirements.

What makes projects successful?

ARC's vice president of consulting, Dave Woll, did a study on this very topic. Although some aspects of the results were to be expected, some were enlightening. Surveyed manufacturers, with a healthy sampling of industry "leaders" and "followers" and a middle category of "competitors," enabled ARC to uncover some important best practices.

Project success best practices:

The degree to which projects are considered to be successful varies widely between the three groupings of industrial companies: leaders, competitors, and followers. Understanding the value of automation also varies widely between the three, and so does the degree to which projects are funded. However, at the leader level, there is a correlation across success, understanding, and funding.

Leaders are more satisfied with their projects. In fact, leading companies generally do not have failed projects. They have systems in place to evaluate the merits of a project up front and then typically employ a stage gate process to reassess and adjust project parameters over time. While overall in the survey 70 percent of projects are considered disappointing or not meeting project justifications, the majority of projects among industry leaders are considered successful. This tells us that with the right processes, the outcome can be guaranteed.

Leaders clearly have a better understanding of the value of automation. When asked to rate their understanding, leaders responded with an equal and high measure of both excellent and good responses. The competitors and followers in the field were nowhere near as confident in their value propositions.

Leaders get more projects funded. Over 80 percent of leader respondents highlighted that more than 60 percent of projects considered are funded. Compare this to the approximately 25 percent of projects successfully funded from other respondents.

In a surprising result, leaders include nonquantifiable benefits, indirect benefits, and life-cycle considerations in their analysis; a significant number of respondents in the competitor and follower category indicated that they did not. Almost all companies include an analysis of total life-cycle cost in their project justification process.

Another surprising result was that leaders spend 50 percent less time performing risk analysis for a project than both competitors and followers. The takeaway here is that they focus on upfront design and then on project management, to assure a quality outcome. Leaders also leverage commercial tools or purchased tools for project evaluation and management, rather than ad-hoc solutions or systems on a case-by-case basis.

All respondents recognized the importance of keeping the same project team for the duration of a project. More than 75 percent responded in the affirmative.

When asked how they apportion the benefits of consultants, financial balance, and rules of thumb, leaders showed an equal respect for each. Other respondents gave a much higher weighting (two to one) to rules of thumb. This likely speaks to the lack of rigor, or goes back to the poor understanding of the automation project benefits when justifying projects. Consultants play a major role in helping leaders justify projects. In fact, according to the survey results, leaders weighed the importance of engaging consultants at over two to one with respect to both best operator advice and rules of thumb.

When reviewing the calculations for economic return, return on investment (ROI) was the most

FAST FORWARD

- Project justifications involve an understanding of the big picture: company competitiveness.
- Project justifications involve building arguments that show an understanding of many business concerns.
- Project justifications involve thinking about alignment with corporate initiatives, not just performance.

predominant method used. Other methods included economic value added, return on invested capital, and return on capital employed. ROI exceeded all other methods combined.

Finally, when surveyed about the involvement of upper management in project justifications, leaders indicated that projects were more successful when management took an active role in the entire process, from project justification to implementation. This likely drives transparency and improved communications to arrive at the desired outcome.

Conclusion

Many factors will affect the success of automation or related information technology projects. But first considerations should be based on overall company directives with respect to its competitive stance. What will make the company more competitive in the future, and how can this project influence that?

Then, it is important for everyone to be involved with the process. Communicate widely throughout your organization. Consider all alternatives and reevaluate and tune the process to drive toward your desired outcome. If your staff is new or inexperienced in making this type of strategic decision, involve industry experts to assist with selection criteria development, architecture development, vendor selection, or cost justification. However, do not delegate the responsibility completely. Make it a team effort.

ABOUT THE AUTHOR



Dick Hill (DHill@arcweb.com) is vice president and general manager of ARC's manufacturing advisory services. Hill is responsible for developing the strategic direction for ARC products, services, and geographical expansion. He is also responsible for covering advanced software business worldwide. In addition, he provides leadership for support of ARC's manufacturing advisory service team and clients.

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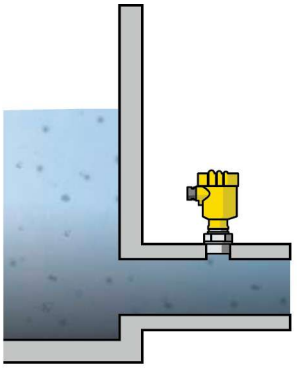
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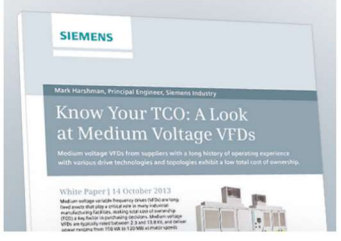
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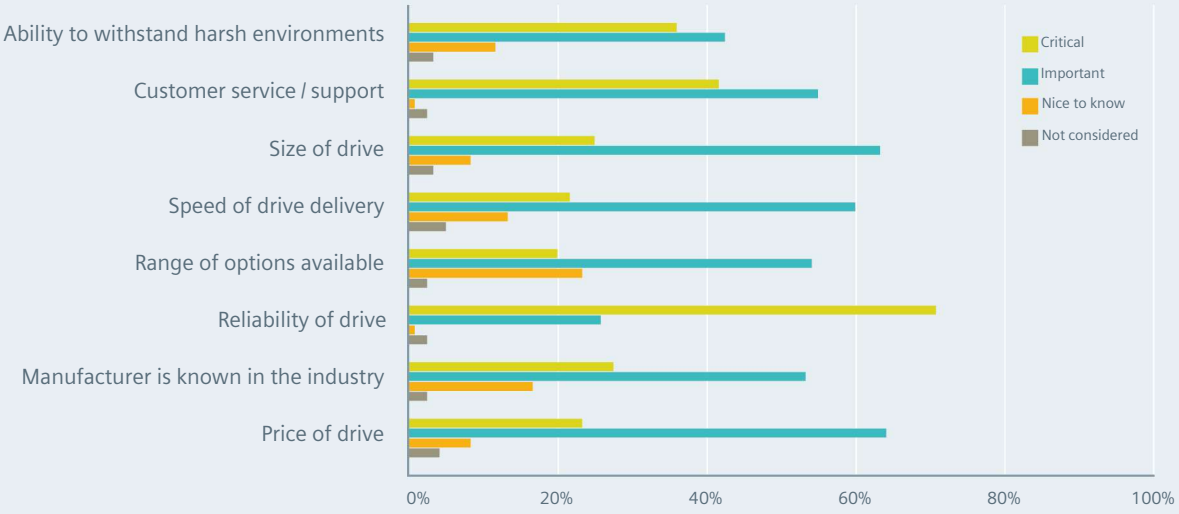
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White Paper: Know Your TCO
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Leading factors considered by purchasers when selecting medium voltage VFDs



Engineering and operations personnel survey conducted by Tritech Marketing Inc. Research.

Know Your TCO

Siemens' new white paper helps manufacturers determine the total cost of ownership (TCO) of medium voltage VFDs.

In today's manufacturing sector, engineers increasingly incorporate medium voltage variable frequency drives (MV VFDs) into the manufacturing process. These high-consideration, high-cost assets deliver a range of benefits, including energy efficiency and improved process control.

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To help today's engineers with the drive selection process, Siemens has developed a new white paper, *Know Your TCO: A Look at Medium Voltage VFDs*. Inside, you'll learn the following:

- Which factors contribute to a drive's TCO
- How to calculate the net present value (NPV)
- The inverse relationship between reliability and TCO
- How product longevity, customer service and support affect TCO

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A Glimpse at TCO's Contributing Factors

The figure above shows the leading factors that purchasers consider when selecting an MV VFD. As you can see, the drive's purchase price is only one of the factors, all of which contribute to TCO. Anyone looking to purchase an MV VFD should explore these factors, as explained in the white paper.

Download the white paper now at usa.siemens.com/TCO



Field wireless networks

By Amit Ajmeri

ISA-100.11a and other wireless technologies are making inroads into process control and measurement applications

Industrial wireless devices and networks are used for measurement and control applications in areas of process plants where it is too difficult or too expensive to hardwire sensors, transmitters, and final control elements. They are also used for temporary applications, such as in research and development and pilot plants. Although most consumer wireless networks are used for convenience, industrial field wireless networks must be much more reliable, and cannot interfere with other wireless applications in the plant. These networks, and their accompanying wireless devices, must also be simple for existing plant personnel to support. Finally, industrial wireless networks and devices must easily integrate with existing wired devices and networks, and the entire wireless system must be flexible and scalable.

Wireless solutions for the process industry

There are three basic areas in which industrial field wireless networks can operate in the process industry: the global canopy, a site backbone, and a field mesh. The *global canopy* is the term used for long-range wireless communication. It can be made up of a site-to-site private network joining locations up to hundreds of miles apart, or it can use public networks, such as the Internet, a satellite, or cellular communications. This type of network is used for data transmission over very long distances.

A *site backbone* is a good solution in cases where data is transmitted from cell-to-cell within a transmission distance of a few miles. Although the distances covered are shorter than with a global canopy, a site backbone network can still be used to transmit data over relatively long distances.

A *field mesh* or wireless sensor network is used for sending and receiving a few kilobytes (kB) of data over a short range up to a few hundred feet. These field wireless networks are comprised of sensors and actuators, field mobile devices, and field end points—and these types of networks will be the focal point of this article.

Frequency bands

Communication frequency is one of the most important factors when implementing a field wireless network. The 2.4-gigahertz (GHz) wireless communication frequency band is most commonly used for industrial applications. It is part of the industrial, scientific, and medical (ISM) radio bands that were originally reserved for international use of radio-frequency energy for ISM purposes, as opposed to telecommunications.

The ISM band has become the de facto standard because it is available worldwide and does not re-

quire licensing. Within that band, 2.4-GHz communications are used in the following standards:

- IEEE 802.11b/g/n (Wi-Fi) is used for a wireless local area network with communication distances of 300 to 900 feet with a data communication rate of a few megabytes per second.
- IEEE 802.15.1 is used for Bluetooth® communications that require extremely low power over a short communication range of 3 to 30 feet.
- IEEE 802.15.4 is a self-organizing, self-healing mesh network used in low-power personal area networks. ZigBee, ISA-100.11.a, and WirelessHART use variations of this standard.
- IEEE 802.16e is used for the WiMax communication protocol and covers a 3- to 30-mile range with a data transmission rate of approximately 72 megabytes per second.

Wireless for industrial automation

Figure 1 depicts various types of networks within a plant. Wireless sensors (XC, XX, XV, etc.) are at the bottom of the diagram. The architecture may include a wireless sensor network working at a remote location exchanging data with the process control network using IEEE 802.11 Wi-Fi or IEEE 802.16 WiMax technologies.

Located at the next level up, between the distributed control system and programmable logic controller automation systems and the wireless sensors, are access points or gateways. Gateways are the interface points between the wireless network and the host automation system. The gateway talks to the sensors wirelessly, and communicates with the automation system using wired protocols, such as serial Modbus, Modbus TCP, or Ethernet IEEE 802.3. With most wired protocols distance is a limiting factor, so the gateways are typically located close to the automation system.

In most industrial applications, there are other devices using wireless communications, such as cameras, radio frequency identification (RFID) systems and cell phones—so methods must be devised to reduce interference and ensure reliable communications.

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- Various wireless topologies can be employed to meet the demands of industrial wireless applications, including star, tree, mesh, and cluster.
- Industrial wireless protocols typically provide security features, such as encryption, authentication, and key management, to ensure protected communications.
- Reliability is assured through redundancy, intelligent channel hopping, time synchronization, and other features.

Network topologies

Wireless network topologies can be used, depending on application requirements, including star, tree, mesh, and cluster. The ISA-100.11a protocol supports all these (figure 2).

The star topology is typically used with reduced function devices that are only powered by batteries. Full function devices that work with batteries or line power are more adaptable for use with other topologies.

The most basic network type is the star topology with one functioning routing or center node that sends and receives information from all the end nodes. Thus, the end nodes only have one function: transmitting data to the center node. As a result, the end nodes consume very little power, because they only send information to the center node, then go back into sleep mode. In addition to reducing the energy needs at end nodes, the star topology prevents any end node failure from affecting the rest of the network, because every end node device is isolated from the larger network by the link that connects it to the center node.

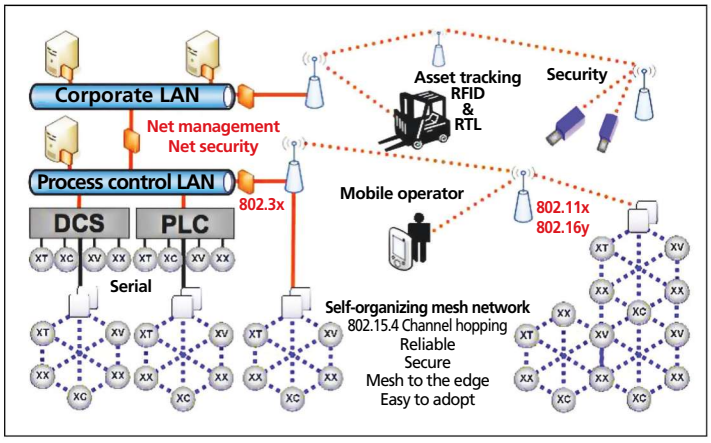
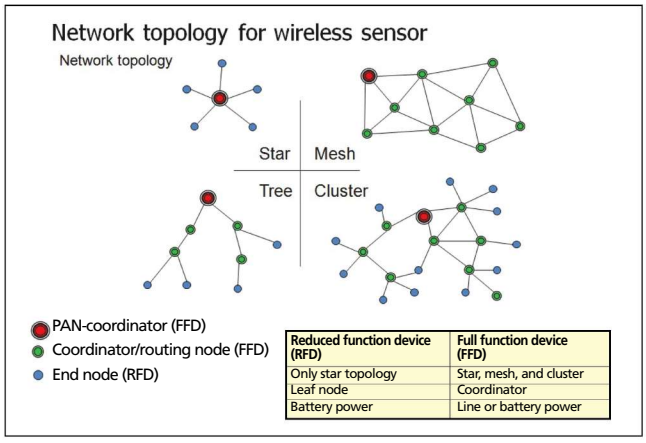


Figure 1. Various wired and wireless network types are often found in a typical manufacturing facility.



Reduced function device (RFD)	Full function device (FFD)
Only star topology	Star, mesh, and cluster
Leaf node	Coordinator
Battery power	Line or battery power

Figure 2. Network topologies for wireless networks can be a single star or tree topology or combinations of topologies, such as a mesh or cluster.

The mesh topology is comprised of nodes that can each route data from neighboring nodes as long as they are in the same radio frequency range. The mesh network system provides reliable, secure data transmissions and is highly scalable. If a path via one node or a set of nodes is not working, a node can route its data to a neighboring node, with the information eventually reaching the destination node via this alternate path.

The tree and cluster topologies are a combination of the star and mesh networks. The tree network has one communication node with star nodes below it, and the cluster network is a combination of star and mesh networks.

Security

Security is a critical factor for industrial wireless systems used in the process industries, so mechanisms for protecting the wireless network must be implemented. The ISA-100.11a standard has several security measures that can keep the network safe, including:

- Encryption: Each ISA-100.11a data communicator has 128-bit encryption.
- Authentication: Only devices authenticated by the system manager and security manager can exchange data.
- Integrity: Each data point uses an end-to-end media access control (MAC) address to ensure data integrity and transport security.
- Key management: All wireless devices must have a join key that acts as a password that the device uses to authenticate it to the network. This is a key differentiator for the ISA-100.11a protocol.

Authentication key management and join-process technology enhance the security of the network by preventing an unauthorized node from joining the network. All node authentications are controlled by the network security manager, which requires any device joining the network to provide all its credentials in an encrypted fashion (join key) to the gateway. Each data packet has a 128-bit encryption. End-to-end basis transport security includes message-level security, such as message encryption, as well as transport-level security, such as Windows.

The network administrator gives all wireless devices a join key and sets the parameters required to access the network. Once the join key is recognized and the device has joined the network, the security manager issues it additional keys (master, session, and private) for further communication. These keys should be periodically updated, because limiting their life spans further protects the network.

The "hop-by-hop" MAC address security requires every data element transmitted between two wireless nodes to provide the MAC address of the originating node as well as the end node for extra security.

Coexistence strategy

Since many wireless standards use the 2.4-GHz frequency band, it is very important that the various wireless technologies (Wi-Fi, WirelessHART, ZigBee, ultrawide band, and others) can operate together on the band. Among the best methods for establishing coexistence among the various wireless communication devices are spectrum spreading, frequency hopping, and time slotted and scheduled transmissions.

Electronic and electrical noise within an industrial plant can cause disturbances if protective measures are not taken. Lower-power radio technology with a spectrum-

Understanding the ISA100 series of standards

ISA100 is a multi-national, multi-industry standards development initiative of ISA, an accredited member organization of the American National Standards Institute. The ISA100 program was launched to create a reliable and universal family of wireless standards and guidelines for industrial automation applications. It is a market-driven (not vendor-driven) standards development program created by a committee that includes end users. The end users actively participate and vote on decisions regarding the standards and on requirements for wireless plantwide communications.

The ISA100 committee is organized into several working groups (WG), including:

- WG14, Trustworthy Wireless, to improve the reliability and security of wireless, as well as to simplify configuration and usage in a plant environment
- WG15, Wireless Backhaul Network, focusing on middle-range cell communications within the plant environment
- WG16, Factory Automation, to cover a completely separate requirement for data latency compared with that of the process industries
- WG18, Power Sources, to improve battery standardization
- WG20, Common Network Management (CNM), to present a CNM framework to monitor and provide actionable information to various and disparate wireless networks commonly found in wireless network environments for industrial automation and control systems. The framework will be scalable to address various network sizes and device populations and extensible to adapt to changing technologies, applications, and user requirements.

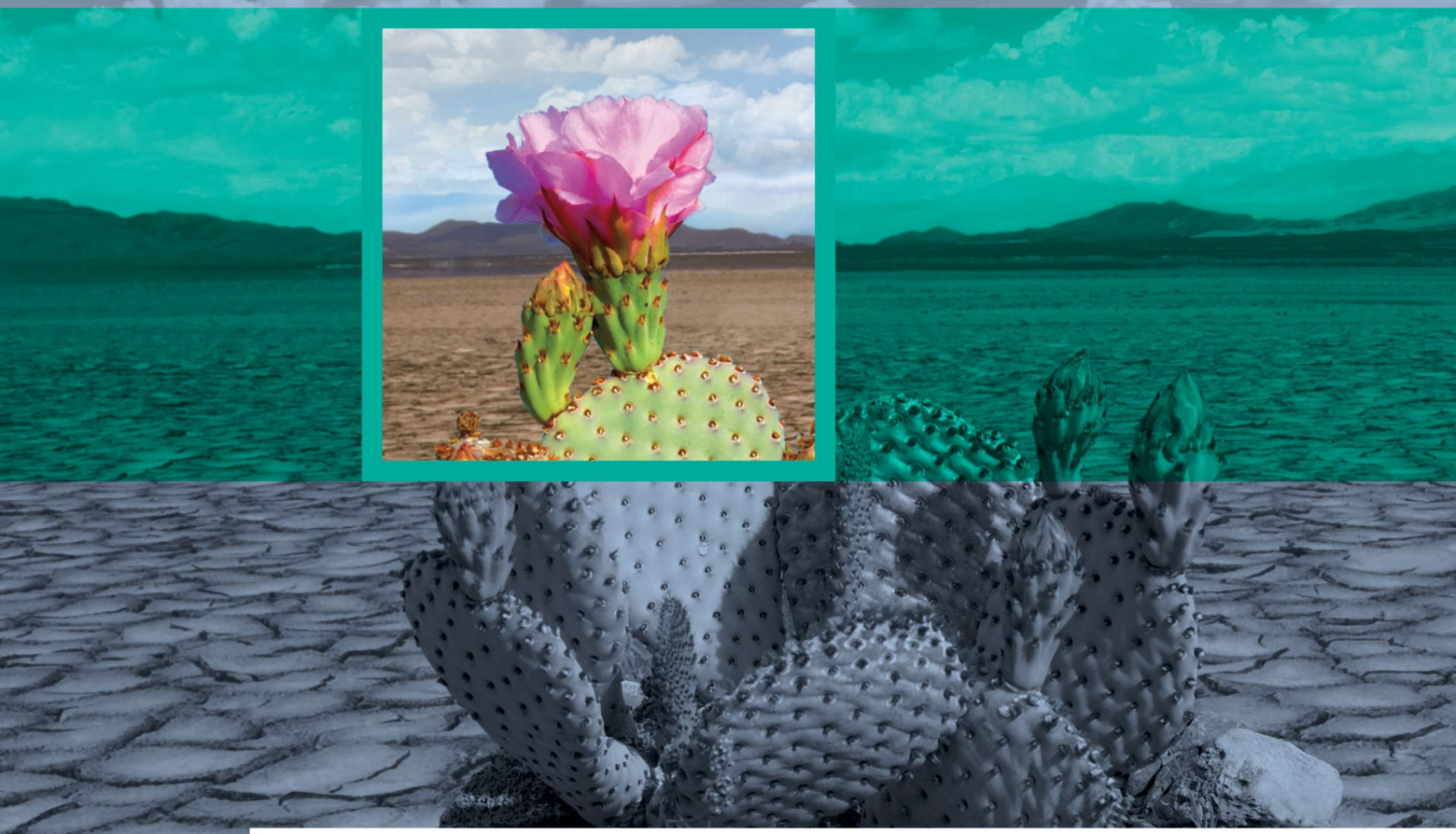
ISA-100.11a-2011 was approved as an ISA and American National Standard in 2011, and is currently nearing final approval by the International Electrotechnical Commission as IEC 62734.

Other documents developed by ISA100 include:

- ISA-TR100.15.01-2012 – Backhaul Architecture Model: Secured Connectivity over Untrusted or Trusted Networks
- ISA-TR100.14.01-Part I-2011 – Trustworthiness in Wireless Industrial Automation: Part 1, Information for End Users and Regulators
- ISA-TR100.00.03-2011 – Wireless User Requirements for Factory Automation
- ISA-TR100.00.02-2009 – The Automation Engineer's Guide to Wireless Technology: Part 2 – A Review of Technologies for Industrial Asset Tracking
- ISA-TR100.00.01-2006 – The Automation Engineer's Guide to Wireless Technology Part 1: The Physics of Radio, a Tutorial

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spread technique deployed at a 200-kB rate helps overcome the noise problem. Data is distributed among various channels, then collected and reassembled by the receiver. Frequency hopping, in which the data rapidly switches among many frequency channels, helps avoid congestion. These techniques are also a good way to increase security, because both the spectrum-spread code and frequency-hopping patterns are necessary to retrieve the data sent over the network.

Synchronized timing enables multiple access capabilities by assigning each device a particular time slot to avoid collisions. Deterministic transmitting (TDMA) can also save power, because only the sending and receiving devices must be awake during the data transmission. TDMA offers synchronized time sense in which each subnet gets time-synched data from the network protocol server, ensuring that all wireless network data transmissions occur at the proper times.

Other methods for overcoming obstacles are multipath mesh networks and intelligent channel hopping. Wireless mesh networks route traffic toward the Internet gateway (IGW), or from the IGW to the access points. When multiple devices attempt to select the best throughput path toward a gateway, the traffic on these paths can diminish the speed and performance of the network. A multipath mesh network seeks alternative paths during times of congestion.

ISA-100.11a also supports the black listing of channels. For example, suppose a plant uses Wi-Fi channel 3 for ISA-100.11a communications, and channels 21 and 24 for general Wi-Fi communications. The system manager can be configured so channels 21 and 24 are blacklisted for ISA-100.11a communications to avoid performance issues on the network. In this scenario, there are two separate channels: one channel for the Wi-Fi and one for ISA-100.11a communications, enabling the two wireless technologies to coexist in the plant.

Reliability

Reliability is of paramount importance for industrial process control and measurement applications, and the ISA-100.11a standard has numerous techniques for ensuring reliable communications such as redundancy, intelligent channel hopping, duocast technology, and time synchronization.

A mesh network offers redundancy, because it can reroute data from one node to the destination mode, avoiding the obstructed node. For additional reliability, redundancy can also be implemented at the gateway, backbone, security system, and system manager. For example, if one backbone router fails, the other router retrieves the data from the sensors, and then sends the information to the gateway.

Another method to ensure accurate throughput is channel hopping, where clear channel access technology dynamically chooses different channels of operation to avoid interference (figure 3). In addition, frequent retry attempts are made to limit data latency to 100 ms or less.

ISA-100.11a also supports duocast for redundant connectivity in which one device sends information to two neighboring nodes simultaneously. The two receiving nodes send a confirmation from both end devices in the same time slot. Without duocast, if one communication path fails, a retry attempt is used before going to a neighboring channel to transmit the data, which can slow communications.

Time synchronization is highly accurate, because each data packet is time stamped using International Atomic Time. All data has a time slot allocation, and the data must reach its end destination and receive a confirmation from each node within that time slot. There is a limited time to capture the data packet and decipher the information, which provides secure communication.

Scalability and flexibility

Each ISA-100.11a end device uses Internet Protocol version 6 (IPv6), the newest Internet technology. This helps "future proof" the network by enabling access from a field wireless node using the latest IP technologies. The ISA-100.11a protocol has many features for scalability in terms of the number of nodes that can be added, as well as how much area can be covered. To expand the network and add more nodes, an additional backbone router can be added to create a subnet. Multiple subnets, slow-hopping mode, protocol mapping, and tunneling and backbone routing are all examples of the multiple functions of the ISA-100.11a architecture.

For fast firmware downloads and increased staff mobility, ISA-100.11a supports the slow-hopping mode using carrier sense multiple access. This enables a channel to be locked for a specific period, instead of hopping every 10 ms. The slow-hopping mode is particularly beneficial when performing a firmware download to upgrade a radio or sensor electronics for a wireless node. It also facilitates using a handheld device for calibration checks, and for configuring a specific device.

Protocol mapping and tunneling reduce infrastructure costs by supporting legacy protocols and combined wired and wireless integrations. The ISA-100.11a architecture supports many existing protocols, including FOUNDATION Fieldbus, HART, Profibus,

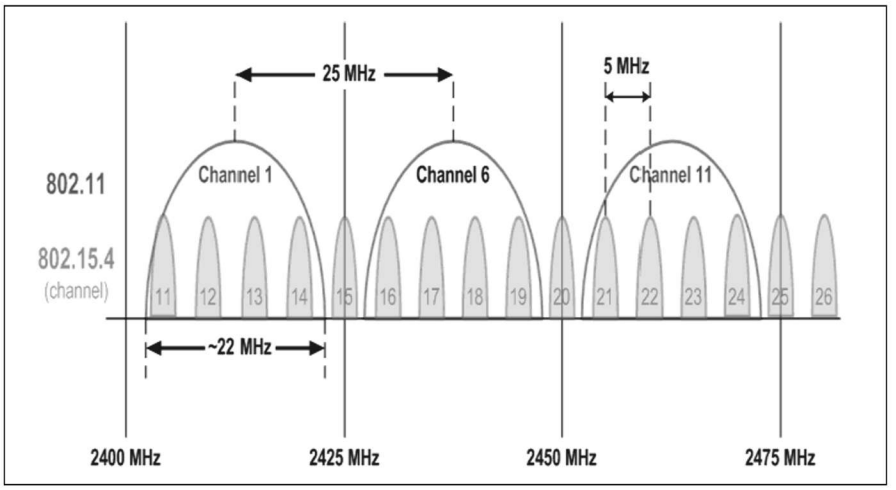


Figure 3. Channel hopping allows wireless devices to dynamically choose different channels of operation to avoid interference.

Modbus, and CIP. The ISA-100.11a stack includes provisions for protocol mapping and tunneling to enable FOUNDATION Fieldbus, HART, and Profibus to use the ISA-100.11a stack for wireless communications. ISA-100.11a is targeted for measurement and control applications that can function with a 1-second data rate, and it also supports peer-to-peer communications for proportional, integral, derivative loop scheduling.

A HART device can transmit data by putting an ISA-100.11a wrapper on top of the HART data, and then sending the data through the wireless network router (figure 4). When the data is received at the gateway, the ISA-100.11a wrapper is removed, and the HART data is sent to the network.

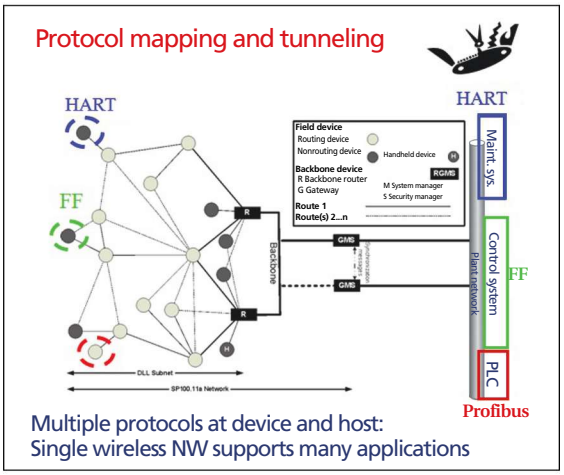


Figure 4. The ISA-100.11a standard can support multiple protocols simultaneously on a single network.

Customizable network performance

To support control applications, a subnet can be created with a 1-second data update rate via direct communications to the backbone router. This enables a 1-second transmission speed from the node to the automation system.

Multiple subnets can work in the same physical space and share a single wireless network with flexible customization for optimal network performance. In addition to sharing a single network, ISA-100.11a also allows networks with different communication speeds to operate in the same physical space. For example, one subnet can be comprised of low-speed sensors, while another provides a 1-second update rate for control applications.

ISA-100.11a supports peer-to-peer control communications, so the data from a measuring device does not need to go to the gateway, but can instead be transmitted directly to the final end control element. These peer-to-peer communications enable data exchange from one wireless node to another within a high-speed subnet for 1-second updates.

An ISA-100.11a wireless network can be deployed and reconfigured through device installation and software configuration and changes—instead of requiring burdensome wiring, addition of I/O modules, programming, and possible upgrades to the automation system. Many of the ISA-100.11a system components, such as the backbone routers and the system manager, are designed as function modules that can be added or removed depending on system requirements.

ISA-100.11a provides reliability, scalability, high performance, and the ability to simultaneously support numerous devices and protocols. Several companies are supporting the ISA-100.11a standard with field instrumentation, automation systems, and software to support the latest wireless technologies from temperature and pressure sensors, gateways, and more.

Interoperability testing and certification from the ISA100 Wireless Compliance Institute (www.isa100wci.org) ensures all manufacturers' wireless devices can communicate with the backbone router and gateway without problems in a plug-and-play manner.

ISA-100.11a wireless network architecture components

Role	Role definition and responsibilities
Input/output	Sources or consumes data. Does not route.
Router	Routes messages for other devices operating in the wireless subnet.
Backbone router	Routes data via the backbone. Mitigates among devices operating in the wireless subnet and devices operating on the backbone.
System manager	The "brains" of the network. Manages all network devices through policy-controlled configurations based on desired performance parameters.
Security manager	Enables, controls, and supervises the secure operation of all devices present in the network.
Gateway	Provides an application interface between the wireless network and the plant network.
Provisioning	Provisions devices with configurations required for operation within the network.
System time source	Responsible for maintaining the master time source of the network.

ABOUT THE AUTHOR

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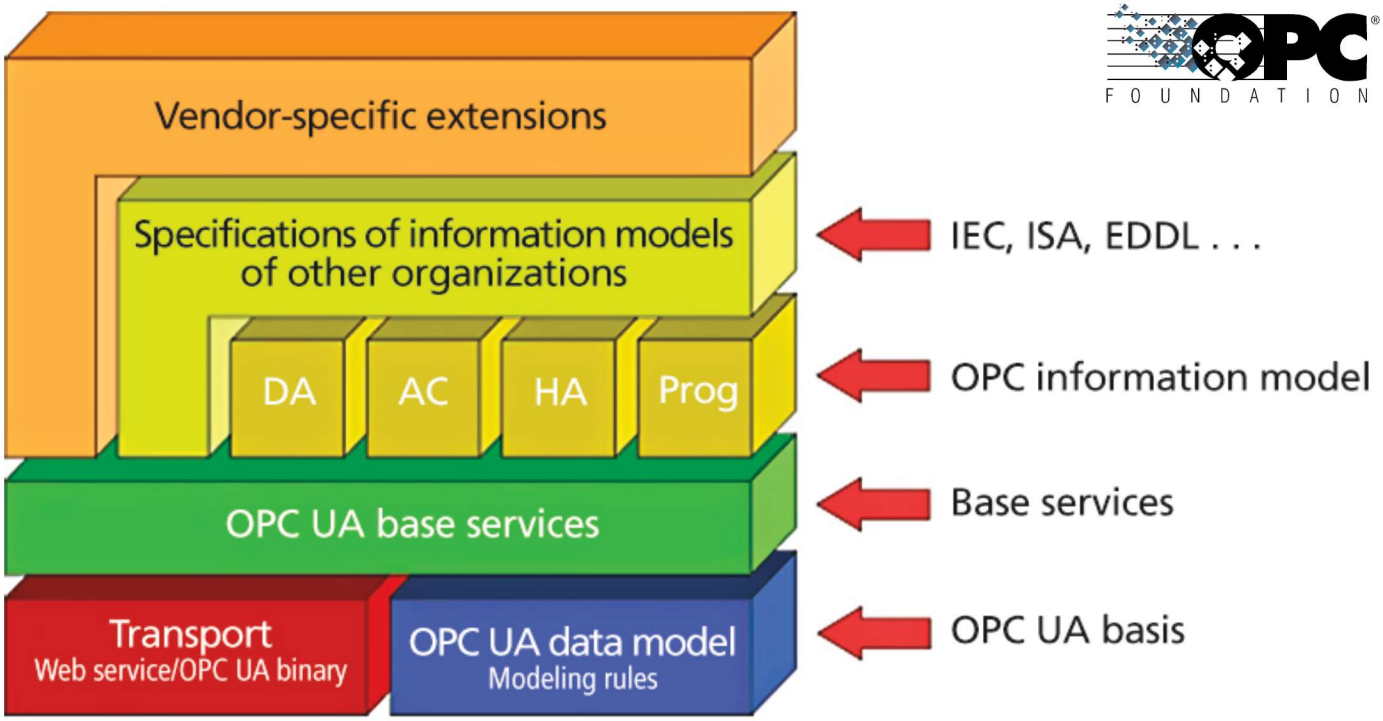
"Analysis of Wireless Industrial Automation Standards: ISA-100.11a and WirelessHART"
www.isa.org/link/analysisofwireless

"A real mesh"
www.isa.org/link/arealmesh

"Opportunities for smart wireless pH, conductivity measurements"
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Wireless Networks for Industrial Automation
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OPC Foundation open standards



Automation and business interoperability improves efficiency

By Thomas J. Burke

Is the concept of open standards all myth and magic? Is there a real business and technical value proposition for open standards like OPC that justifies the end users demanding the suppliers to deliver? How does OPC provide timeless durability and deliver specifications and technology that the suppliers build into their product portfolios? This article explores the OPC vision, and the who, what, when, where, and why of OPC interoperability in 2014.

OPC became the de facto standard for data acquisition for discrete and process control applications, and over time has evolved and revolutionized industrial automation multiplatform interoperability. The OPC Foundation strategy is providing the infrastructure to enable the community of suppliers, consortia, and end users to collaborate to bring interoperable products to market. In addition, the OPC Foundation had established a certification program that validates the interoperability of products to meet and exceed the quality standards of the end user community.

What are the benefits?

Typically suppliers have always preferred the brand-loyalty architecture; we want our end users to only use products from "our" company and our preferred suppliers. The suppliers assert and assume they know what their end users want and need! Suppliers always tell the story that standards are the least common denominator approach, and they can give users better compatibility and performance along with guaranteed interoperability with their proprietary architectures.

The world of consumer electronics and the expectations set with out-of-the-box multivendor plug-and-play interoperability is changing the world. End users have high expectations that technology innovations in consumer electronics must be propagated to industrial auto-

mation, and that tablets and smartphones must be integrated with their industrial automation devices and applications.

OPC Foundation

Has the OPC Foundation successfully delivered the right technology and value proposition to change the landscape of industrial automation and related domains?

OPC was started in 1995 by a group of companies that had the vision for a simple interface that allowed first-tier visualization applications on Microsoft platforms to read and write data from process control and factory automation devices. The value proposition at that time was simple; vendors were looking to make their lives simpler and more cost-effective. They did not want to be burdened with continual development of complex device-driver software to talk to the myriad devices. Before OPC, companies had developed their own "proprietary open" technology and distributed tool kits to multitudes of vendors to develop software as an interface for their software to discrete and process devices.

When OPC first started, it was all about a small set of vendors. It was a community thinking and collaborating to solve a critical problem of moving data from the factory and process control devices to the first-tier visualization applications. The key concept here is a community collaborating together.

OPC continued to develop new standards beyond the first data-access specification that was developed in the mid-1990s. OPC solves problems beyond data access, including specifications for alarming and historical data access. The Microsoft platform-based specifications of data access, alarms of events, and historical data access are now labeled the classic OPC specifications.

OPC Unified Architecture

Requirements and technology continued to advance, which established the need for a new architecture that addressed a far broader scope of interoperability problems.

In 2004, the OPC Foundation took on the task of developing this new architecture called the OPC Unified Architecture (OPC UA), which provides a service-oriented architecture designed to provide multivendor, multiplatform, secure, and reliable interoperability. The OPC Unified Architecture unifies all the data and information models from the classic OPC specifications. It provides an extensible information model that allows not only OPC classic data to be modeled, but also allows other consortia and vendors to render their respective information models into an OPC UA "name space." This unique information model architecture allows applications to exchange data/information with applications and devices, without having advance knowledge of the "data structures."

FAST FORWARD

- Is there a true business and technical value proposition for open standards like OPC?
- Multivendor interoperability at its finest leverages the capabilities of the information model and OPC UA.
- The business value proposition for OPC is frictionless information integration and lower total cost of ownership.

The OPC Foundation membership is the community that continues to collaborate to develop the best specifications, technologies, certifications, and processes. Years ago, the focus was exchanging data inside a corporate firewall. No one "really" worried or thought too much about security. No one thought about the volumes of data and what to do with the volumes of data. Yesterday and today, it is all about converting terabytes of data into intelligent information. After OPC released OPC Data Access, the OPC Foundation continued to work with the community of collaborating vendors to look for other interoperable problems. Some of the specifications that resulted solve the problems of alarming and historical data access.

Classic OPC is the "base" of the OPC Unified Architecture. OPC UA expands classic OPC from the embedded world to the enterprise. It is all about connectivity in and between industrial automation and all the related domains.

OpenO&M

One of the first OPC Foundation collaborations was OpenO&M (operations and maintenance), which was a partnership between organizations including MIMOSA, ISA95, OAGIS, and OPC. It was based on the initial work between OPC and MIMOSA, focusing on asset management and the idea of preventative maintenance and reliability applications. MIMOSA really provided the deep repository of information, and OPC became the mechanism for transporting the information between various applications.

Broadening scope

In the past two years, the focus has been elevating the collaboration with other consortia and capitalizing on the information modeling concepts inside of the OPC unified architecture. The focus of the collaborations has spanned oil and gas, building automation, smart energy, and machine-to-machine connectivity. Some of the specific consortia that the OPC Foundation

is collaborating with include MDIS, DSATS, MTConnect, PLCOpen, BACNet interest group Europe, Energistics SLC, ISA95, and ODVA/Sercos.

So how does this collaboration architecture work? First, OPC has what is known as the information model architecture. The OPC classic specifications, including data access, alarms and conditions, and historical data access, are modeled as information inside the OPC name space. OPC UA took the classic OPC specifications and separated out the services from the information. This enabled us to identify the key services of reading, writing, browsing, subscription, and security and to keep them generic from the information that they operate on. We are also able to provide complete structures of information behind the data, which can be thought of as the data and then the metadata behind the data.

OPC provided its own information model for its various classic OPC specifications. Then the OPC Foundation looked at the opportunity to model information from other vendors and other consortia's information in a generic way.

The best way to explain the information model is to show a picture and show the separation between the OPC UA–based services and the OPC information model and the specifications of the information model from other collaborative organizations.

Collaborations

In the oil and gas community, OPC is engaged in a collaboration called MDIS, which stands for master control system (MCS) – distributed control system (DCS) interface standardization (IS). All the major oil companies (the operators), all the major DCS vendors, and all the subsea vendors collaborated to standardize the objects for automation from the subsea world to the DCS. They wanted to standardize on their communication protocol and information model, with the benefits being reduction in cost and simplifying engineering. The key features that they will use from OPC UA are subscription, security, robust medication, and redundancy—but most importantly leveraging the information-modeling capabilities, including all of the subtyping of information. The architecture possibilities include connectivity to the subsea vendor hardware through a subsea gateway that is topside to the MCS and ultimately to the DCS human-machine interface (HMI) system. Standardizing information will allow all the equipment to operate seamlessly and talk the same semantics and syntax. Multivendor interoperability at its finest leverages the capabilities of the information model and OPC UA. And an OPC UA companion specification is being developed that will describe all the objects inside the system and all the information necessary to provide the infrastructure for interoperability and compliance.

In other areas of oil and gas this is known as drilling system automation (DSA-TS). This is all about looking at the surface equipment of a drilling rig. How do you model all the objects and integrate information between multiple vendor systems? This is collaboration between all the operators, the drilling contractors, the equipment suppliers, and the service companies that essentially are cooperative adversaries interested in information modeling and interoperability. The companies that work together will ultimately reap the benefits of OPC UA. There will be commercial off-the-shelf products that will easily support the complex information models, providing high degrees of maintainability in a secure fashion. One of the key standards is known as the well (oil and gas well) information transfer standard, which will be in the OPC UA information model and is used throughout the drilling industry.

The companies that work together will ultimately reap the benefits of OPC UA.

Finally, in the oil and gas world is the Standard Leadership Council. The OPC Foundation is proud to be engaged in this important initiative where all the upstream oil and gas consortia are cooperating. One of the specific initiatives coming from this particular effort is the integration between OPC and Energistics. We are putting together a working group that will develop an OPC UA companion specification for PRODML.

Outside of oil and gas we have been working with the building automation community. Specifically, the OPC Foundation is engaged with the BACnet interest group Europe, which is promoting BACnet in the European markets. BACnet is a data communication protocol for building automation under the auspices of the American Society of Heating, Refrigerating, and Air-Conditioning Engineers. It is considered the global standard for communication in building automation systems. In the fall of 2012, the OPC Foundation and the BACnet interest group formed a working group with the main task of creating an information-mapping model for OPC UA and BACnet. Excellent work is being accomplished where we are taking data from building automation inclusive of energy-metering data, real-time operating information, and maintenance information to simplify the integration with enterprise systems, including accounting, operational surveillance, and extensive energy-monitoring applications. In addition, we are able to take data from automation systems inclusive of the production information programmable logic controller (PLC) and DCS used in building automation and other energy supply systems and integrate them with historical data storage and scheduling applications.

Embedded

OPC UA was initially focused on providing the infrastructure necessary for data and information integration between the control systems in the information technology world. As OPC began to provide the necessary services with respect to security and the corresponding scaling of services for platform independence, we

recognize a unique opportunity for OPC UA to actually be scaled down into the embedded world. OPC UA has now been ported and made available on the smallest level chips, and at the same time OPC UA is now available in a variety of PLC and DCS systems, and numerous vendors are building OPC UA into embedded HMI devices. More importantly people are starting to look at OPC UA as a mechanism for data integration down into the instrument level. OPC UA can exist anywhere. This existence anywhere will allow the integration between industrial automation and multiple related domains.

With respect to the embedded world and collaboration, we have an excellent collaboration between OPC and PLCOpen. In 2009, the initial plan was put together to map data structures inside of IEC 61131 into an OPC UA name space, and the initial goal was completed by March 2010. In 2000, we began the development of PLCOpen OPC UA function blocks and services. The OPC client now resides inside of the PLC and is able to interact through programmatic method calls into the manufacturing execution system world, which behaves as an OPC UA server running in the cloud. The applications of this are very interesting, and we now have the ability to make transactional calls from a PLC leveraging the collaboration between OPC and PLCOpen.

Collaboration journey

The collaboration examples between the OPC Foundation and other organizations are just the beginning in the journey to standardize information integration.

The value proposition of open standards recognized in the world of consumer electronics is gaining momentum in industrial automation. Expectations are clear from the engineers of tomorrow—they rely on multivendor plug-and-play interoperability in consumer electronics and expect multivendor plug-and-play interoperability from industrial automation. More importantly, the ability for all the standards organizations to work together in a harmonization of those information model standards is being realized through the organizational collaboration taking place across our industry.

The multitude of members inside the OPC community are bringing forward products and services that take advantage of the multiplatform architecture from the world of consumer electronics and entertainment. The smartphone has become a way of life. People use their smartphones as a replacement for global positioning systems. People use their smartphones for bidirectional information exchange. Smartphones have become many people's computers. The proliferation of applications and the importance of applications being developed that tie into the world of industrial automation can now be realized in a standard way by leveraging the OPC UA technology.

In summary, the business value proposition for open standards and specifically OPC is really all about looking at the total cost of ownership, and how products from multiple vendors can have frictionless information integration out of the box with plug-and-play interoperability. Consumer electronics are driving expectations for industrial automation end users and engineers of tomorrow. Frictionless interoperability can now be realized by all our suppliers developing best-of-breed products that leverage open standards inclusive of the OPC technology portfolio.

ABOUT THE AUTHOR

Thomas J. Burke (thomas.burke@opcfoundation.org) is the OPC Foundation president and executive director. He was one of the original architects of OPC, and his vision of interoperability is to develop and maintain the best technology, specifications, certification, and process to enable multivendor, multiplatform, secure, and reliable interoperability for and between industrial automation and related domains (www.OPCFoundation.org).

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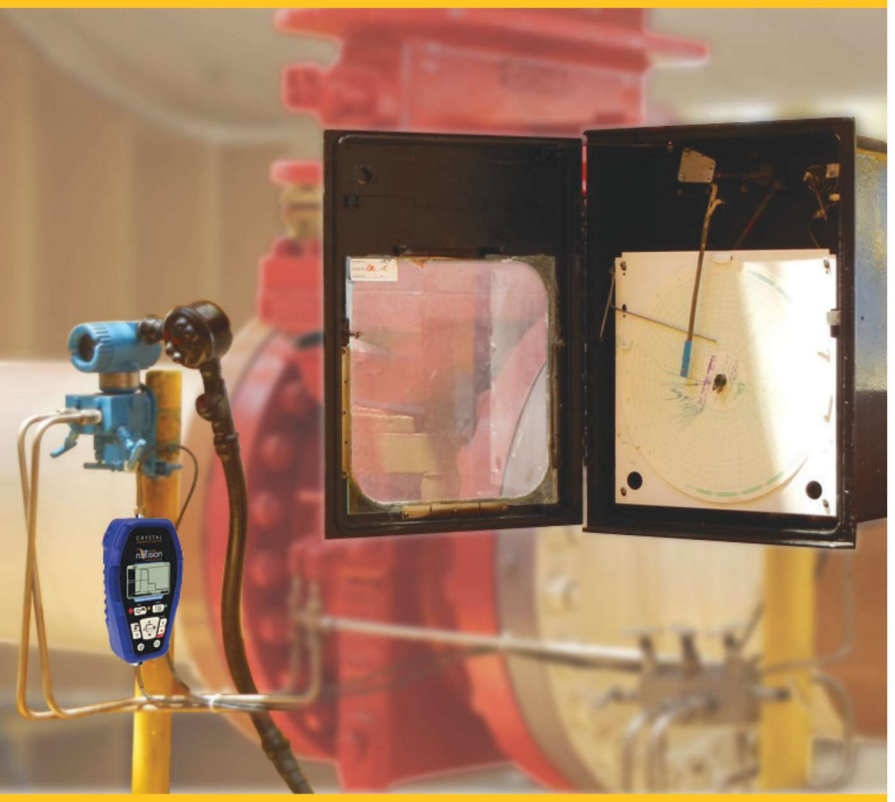
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Reference recorders shake up old standards



Reference recorders offer a reliable improvement over the decades-old technology commonly used.

Reference recorders are a new class of instrument, displacing common pressure instruments

By David K. Porter, P.E.

Within the pressure measurement industry, relentless innovation has become the norm. New technology continually supplants the old. In other technology spheres, PCs pushed aside typewriters. Cell phones drove out telephones. And today, reference recorders are shaking up old standards in the oil and gas industry.

A reference recorder is a new class of instrument that is gaining ground over other common pressure instruments. Chart recorders, deadweight testers, and analog pressure gauges all brought several advantages in their time. Increasingly, reference recorders replace all three with a single instrument.

What is a reference recorder?

Reference recorders unify the recording capability of a chart recorder, the high accuracy of a deadweight tester, and the continuous measurements of a pressure gauge into one tool that is easier to use and less expensive.

On their own, none of these features are unique, but when you put them all in the same device that reduces cost, it is something really useful. These reference recorders are effective for both laboratory and field applications. While data logging devices are nothing new, improved microprocessors allow today's reference recorders to read and record faster and more accurately than ever.

Reference recorders detect readings from modules, which may read pressure, temperature, current, or voltage, and store the data in their digital memory. These modules are usually interchangeable, and recorders generally accept two inputs at a time. In addition to their compact size, reference recorders offer several improvements to bulky chart recorders.

Improving on a chart recorder

Chart recorders, originally patented in 1915, produce a graph on a moving paper chart. They rely on ink pens mounted on mechanical arms, which pivot in response to pressure. Their primary advantage is their ability to record for

FAST FORWARD

- Reference recorders improve on decades-old technology used in chart recorders.
- Reference recorders offer deadweight tester accuracy in a small field-designed instrument.
- Reference recorders remove the need to carry multiple gauges by replacing them all with one tool.

extended periods in remote locations. Chart recorders may be battery powered or entirely mechanical (requiring no external power), allowing them to operate in hazardous locations.

An experienced user may provide preliminary analysis of the output from a chart recorder in the field. However, communicating or storing this data electronically becomes possible only after a technician enters or scans the chart into a computer. This process typically takes several days. Chart recorders usually claim an accuracy ranging from 0.25 percent to 1 percent of span, but that figure depends on changes in ambient temperature and on the thickness of its pens—which can cause an additional error up to 1 percent of the recorded reading. Chart recorders are also susceptible to additional errors caused by overpressure and can easily be damaged if dropped.

By contrast, a reference recorder exports digital data in a familiar spreadsheet that can be easily shared and understood by multiple users. Accuracy remains consistent with temperature changes, is protected from high overpressure events, and is not affected if the gauge is dropped. Despite being battery powered, reference recorders are also rated intrinsically safe for use in hazardous locations.

New regulations in many industries require that safety tests be documented in an electronic format. Some even require that the files produced are in a tamperproof, secure file format. Reference recorders can save the digital data in a signed pdf file, which produces read-only files that cannot be manipulated.

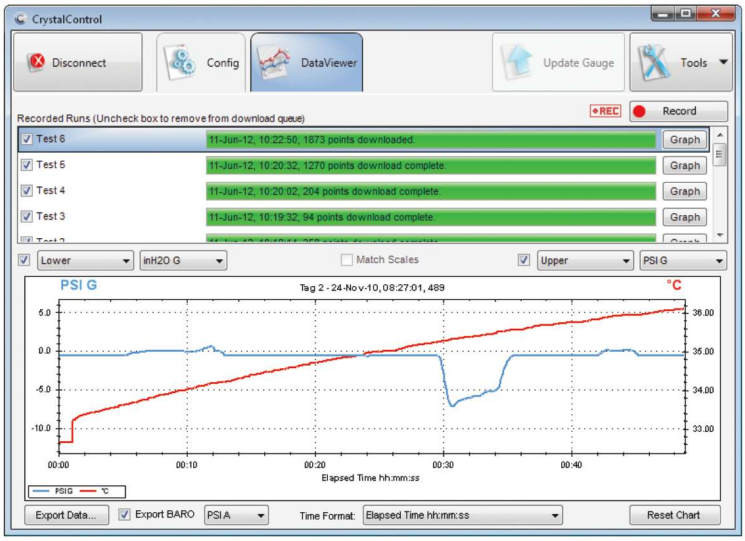
Using deadweight testers in the field

Deadweight testers are used primarily in a laboratory setting. They produce a repeatable reference pressure by balancing calibrated masses on a piston of known area. Operators must use a different set of masses for each engineering unit (i.e., one set of weights for pounds per square inch [psi], another for kilopascal [kPa]). In an indoor, temperature-regulated environment, deadweight testers can be very accurate—serving as a primary standard for the calibration of other pressure measurement devices. Field versions also exist but, outside the lab, their accuracy depends on a host of factors.

To deliver an accurate reading, a field deadweight tester must be perfectly level. Additionally, ambient temperature, humidity, wind, and local gravity will all affect the reading. Without adjusting for these, a field deadweight tester's accuracy may be no better than a common pressure gauge. Even with these compensations, a field deadweight tester cannot actively detect or record changes in pressure; it only offers an accurate reference pressure for comparison.

A reference recorder serves two purposes. First, it stores readings continually, with update rates up to 10 readings per second, and allows the operator to record an entire test from start to finish; second, it provides a consistent, accurate reference for field calibrations. Measurements from modern reference recorders are independent of local gravity or humidity and produce their accuracy via a proprietary mathematical algorithm that corrects for ambient temperature. Achievable accuracies range from 0.025 percent of the indicated reading at lower pressures, to 0.1 percent of reading up to 15,000 psi.

A good reference recorder's accuracy should



Any PC can download the digital data from a reference recorder.



Reference recorders unify the recording capability of a chart recorder, the high accuracy of a deadweight tester, and the continuous measurements of a pressure gauge into one tool.

not change with temperature across its operating range. This is where recorders vary widely. The best manufacturers will provide proof of their accuracy claims, with a calibration certificate from an accredited lab, showing test data at different temperatures.

Linearity and ruggedness are the two most important features of any pressure gauge.



Replacing multiple pressure ranges

Pressure and temperature gauges are widespread and varied in form. Gauges will continue being useful for years, because of their portability, customizability, and affordability. In contrast, their drawbacks stem from their inherent fragility and the necessity to carry multiple gauges to cover a given pressure range.

Linearity and ruggedness are the two most important features of any pressure gauge. Many pressure gauges would be useful across a much larger pressure range if their sensors were more linear and predictable. Also, many gauges continue to display readings that appear accurate, even after their sensors sustained damage.

To overcome these challenges, reference recorders extend their usability by accepting different modules, each of which covers a wide pressure range. Within each module range, a reference recorder will have superior accuracy and linearity, compared to most pressure gauges. The advanced self-diagnostics in a reference recorder prevents the device from displaying an invalid reading if it sustains damage.

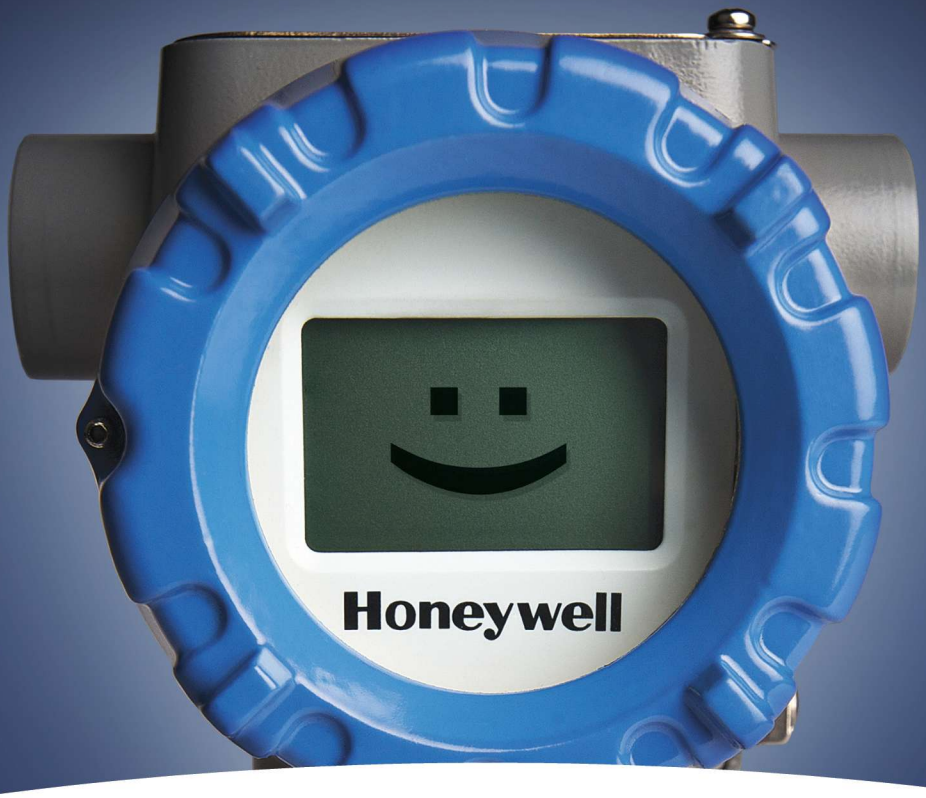
Reference recorders offer a reliable improvement over the decades-old technology commonly seen in the oil and gas industry. Replacing chart recorders, deadweight testers, and older pressure gauges with reference recorders helps companies reduce their equipment costs, training time, and operator error.

ABOUT THE AUTHOR

David K. Porter, P.E., is the division vice president, Crystal business manager at AMETEK. He has more than 25 years of experience; the last 13 years were spent in the test and measurement field developing rugged, highly accurate, and easy-to-use instruments. He has a degree in mechanical engineering from California Polytechnic State University, San Luis Obispo, where he is still active on the Industry Advisory Board. For questions, email sales@crystalengineering.net.

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Remember the customer

By Jon Olson



How many times have you decided to stop working with a company because you thought it was not listening to you? Perhaps you have switched your bank, Internet service, or cell phone provider, because you found it impossible to reach someone who seemed even slightly interested in solving your problem.

You are not alone. Research shows that 68 percent of customers stop working with a company due to treatment they received or perceived indifference. Only 9 percent leave due to competitive reasons.

Today we have great communications and collaboration tools that enable us to make a technical connection almost immediately; we can communicate more quickly across greater distances than at any other point in human history. However, we need to remember that all manufacturing, and other types of business, are primarily driven by people and relationships. Maintaining a personal touch with our customers is what really matters.

Our number one mission must be to help our customers accomplish what they have hired us to do—make their jobs easier and their operations more efficient. Period. Those who ignore this do so at the risk of losing business.

When was the last time you had a meaningful, productive discussion with one of your customers? One that resulted in a real path forward to solve a problem, or better yet, looked ahead to the customer's short- and long-term goals? One that discussed different options to reach these—and the expectations you both have from your working relationship? All too often, conversations with our customers are initiated as a reaction to an immediate problem or issue, rather than as a collaborative step forward. It is too easy to hide behind the ability to "communicate" quickly. Just because you can reach your customer immediately using today's technology, does not mean that you have really connected.

What can we do to improve how we work together and what we deliver, ensuring we develop the products and services that will help our customers use the latest technologies to make their processes more efficient, and their companies more competitive?

It takes a commitment to focus on the people behind the business relationship—and to learn what they need to be successful in their jobs and in the marketplace. It involves taking the time to know the individuals whom you are working with and understand their day-to-day challenges. What is keeping them up at night, and why? What is their "wish list" of solutions that would make their jobs easier? How can you help?

When you actively engage your customers and learn about their challenges, you can anticipate what they will need. This is achieved by building a relationship with your customers, investing time in a two-way dialogue, and really listening to what they say.

A lot of this seems to be common sense, but as we hurry to meet our deadlines, and just quickly touch base with our smart phones or tablets to "close the loop," how many times do we pause and really hear what our customers are telling us? Electronic message exchanges are fine for a quick status update or question, but it is difficult to get to the heart of any problem with electronic communication alone. You can accomplish far more over a cup of coffee, lunch, or even a phone conversation than you can with countless emails or text messages, many of which may not be read.

That is because people are what drive business, and the communication methods are only an effective enabler if you have established a sound relationship first. We need to remember that decisions to proceed with a project, implement a new technology, expand a business, or stay with or switch vendors, are all made by people, and not faceless entities.

As we all rush forward to develop and find new ways to leverage the best new technologies and solutions, we need to keep our focus on the people whom we are creating these for. What tools do they need, and what support do they need from us? How can we deliver these more effectively? How can we make their work environment more efficient, safe, and productive, and their jobs more rewarding? Are we listening to their input and looking ahead to what they will need next?

The time spent to listen and learn about your customers' daily challenges, short- and long-term goals, and priorities will be one of the best investments you can ever make for your business.

ABOUT THE AUTHOR

Jon Olson is North America region division manager, process automation, for ABB Inc. The division develops, produces, sells, delivers, and services products and solutions to enable industrial and energy customers to operate more efficiently and profitably. Before his current role, he served as the division manager, process automation, for ABB in Australia.

Four key benefits of integrated safety control networks

By Jay Cross, CAP

There has always been some form of safety systems implemented in industrial manufacturing automation. However, the manually driven, hardwired safety systems of the past are losing their place in the modernized plant to networked safety systems. This is a new approach; in the past five to six years you would not have seen these integrated systems in applications.

Today, the trend is to make safety more methodical and more integrated into the core of the control systems—not just added as an afterthought. Integrating safety systems into the line control systems at the plant floor is now a part of the safety system design and a requirement in the electrical specifications of most of our clients. We certainly encourage all customers to follow this methodology.

In the old-style architecture, there were hardwired safety systems that were functional, but they lacked key features and functions to protect against possible failure, creating an unsafe situation. For example, there is now redundancy in the devices, the mechanics, the electronics, and the software and hardware. Now, we can centrally manage all the safety devices, such as push-button E-stops, pull cords, and light curtains. They are integrated directly into the line control system and programmed for flexibility to easily control these devices in zones while maintaining safety ratings.

The safety system converges with the line control systems and is held separately and managed based on safety-level requirements. This allows the safety system to be validated separately and to provide plantwide consistency. Key benefits of an integrated system include the following:

Tamperproof

Some operators tend to use the E-stop as a convenient method to stop the machine. E-stops are designed for emergencies, but are used in nonemergency scenarios all too often to stop the line. This generally causes longer downtime and affects production related to restarting the machines.

An integrated safety system allows plants to create safety zones to control how the line is stopped or the areas of machines that are safe to access to clear jams without stopping the machines or the entire line. Now criteria can be developed and programmed into the integrated safety software for safe access and to maintain productivity on the line. Before, those changes had to be made with hardwiring and machine changes.

In general, the Occupation Safety and Health Administration (OSHA) requires the employer to provide a safe work environment and to use a proven method to assure that one exists.

Control

Disparate controls and data slow down operations. By integrating all of the safety controls, plant managers and operators can easily view the data, identify the problems, and decide the best course of action. Diagnostics and alarms can be designed from collected safety information. The customers determine how they want to monitor the system and what data they want to see, such as downtime monitoring.

Before, it was nice to have data; now it is required to have data. Supervisory software can drill down to interrogate each device. This gives the engineering staff the data to access the device and monitor the equipment for failures and frequent incidents. They can make adjustments to improve overall equipment effectiveness. The fault codes provided tell engineering exactly what happened. Our objective is to provide a data-rich environment.

Flexibility

Flexibility on a line has become increasingly important for manufacturers to meet changing customer demands. Whereas hardwired safety systems were analogous to casting concrete or carving stone, digitally integrated safety systems are designed to be nimble. Standardized design allows for changes on the line and expansion efficiencies. The installations typically only require a couple wires, and new parts and old parts can be swapped out in minutes.

OSHA/ISO-13849-1

In general, the Occupation Safety and Health Administration (OSHA) requires the employer to provide a safe work environment and to use a proven method to assure that one exists. Therefore OSHA relies on ISO-13489-1 to define what makes an integrated safety system safe.

The ISO-13849-1 standards help define the safety (performance) level, the safety functions required, the safety control system to be used, and the safety devices to be used. This is all done by identifying the hazards defined in a risk assessment.

By understanding these requirements, designing and integrating the system to these standards, and properly documenting the safety system, customers can be assured that OSHA requirements are met and that a safe system has been integrated.

ABOUT THE AUTHOR

Jay Cross, CAP, is project lead at Polytron Inc., a certified member of the Control System Integrators Association (CSIA) with headquarters in Duluth, Ga. Contact info@polytron.com or visit the website at www.polytron.com. For more information about CSIA and CSIA certification, visit www.controlsys.org.

ISA Certified Automation Professional (CAP) program

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

CAP question

If a hacker intercepts and changes set point data traveling over an industrial network, which basic security property is affected?
A. integrity
B. functionality
C. availability
D. defensibility

CAP answer

The correct answer is **A**, integrity. Data integrity implies that the data received is the same (value, format, quality) as the data sent. If a hacker is successful in changing set point data as that data travels over the network, the hacker has compromised integrity of the data, since it is no longer the same when received as when sent.

Answer B is incorrect, because functionality is not a basic security property.

Answer C is incorrect, because the problem statement did not address the availability of data. It appears that only the value of the set point (data integrity) was affected.

Answer D is incorrect, because defensibility is not a basic security property, but rather a measure of the vulnerability of a system.

Reference: Trevathan, Vernon L., *A Guide to the Automation Body of Knowledge, Second Edition,* ISA, 2006.

ISA Certified Control Systems Technician (CCST) program

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

CCST question

Which of the following flow measurement devices does not require square root extraction?
A. magnetic flowmeter
B. venturi flowmeter
C. orifice plate
D. pitot tube

CCST answer

The correct answer is **A**, magnetic flowmeter. Square root extraction is a mathematical method that is required for flowmeter readings where a measured differential pressure, or pressure drop, is used to infer the flow rate. From the choices in the problem statement, the magnetic flowmeter is the only sensor on the list that does not depend upon a differential pressure to be developed to quantify flow rate, and therefore, does not require square root extraction.

Answer B, venturi flowmeter, infers flow rate by measurement of the differential pressure across a tapered tube.

Answer C, orifice plate, infers flow rate by measurement of the differential pressure across a thin plate with an orifice bored through it.

Answer D, pitot tube, infers flow rate by measurement of the differential pressure between a reference pressure (static pressure of the flowing fluid) and the pressure measured at the impact tube.

Reference: Goettsche, L. D. (Editor), *Maintenance of Instruments and Systems, Second Edition,* ISA, 2005.

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ISA salutes our partners.

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Thermocouples versus RTDs

Choosing the right temperature sensor requires evaluation of the environment, temperature range, accuracy, and speed of response

By Ehren Kiker

Thermocouples (TCs) and resistance temperature detectors (RTDs) are the most widely used temperature sensors in automation and process control. They are found embedded in motors, valves, turbines, bearings, and a host of other devices. Most smart instruments such as flowmeters, pressure transmitters, and level transmitters also have an embedded temperature sensor—used to correct the primary measured variable or for process control.

When used alone, temperature sensors are often installed in thermowells, which are inserted into tanks, vessels, and pipes. A thermowell protects the sensor from the environment, but it slows the response time and degrades the accuracy. Installing sensors in a thermowell is a different subject, and will not be covered here. Instead, this article discusses direct immersion sensors; i.e., TCs and RTDs that are directly inserted and exposed to the process without the protection of a thermowell.

Theories of operation

A *thermocouple* consists of two wires of dissimilar metals, joined at both ends—at the reference point outside the process (cold junction) and at a junction at the point of measurement (hot junction). The metals react differently to temperature changes and generate an electromotive force (EMF) voltage based on the temperature differential between the junctions (the Seebeck effect). A *resistance temperature detector* is based on the principle that electrical resistance in a wire increases with temperature.

In both cases, the sensor is wired to a transducer or signal conditioner that has been calibrated to accept the input voltage or resistance, calculate the correct temperature, and output it as a 4–20 mA, mV, or digital signal to an automation system.

The above is pretty basic, taught in every Instrumentation 101 class. But it poses the first question to consider when choosing a sensor: How do you wire the sensor to the transducer, signal conditioner, or automation system? Such devices and systems, being electronic in nature, need to be mounted in a reasonably safe location, away from high temperatures.

TCs must be wired with thermocouple extension wire, which is the same as the wire used in the TC. For example, a Type K TC uses a wire of nickel-chromium connected at the sensing junction to a wire of nickel-alumel. Extension wire must be the same composition, that is, one nickel-chromium and one nickel-alumel wire. In general, longer runs of extension wire are discouraged, as the wires act as an antenna, making the measurement more susceptible to electromagnetic and radio frequency interference. Cost may also be an issue when dealing with long extension wires, especially ones with exotic materials (e.g., Type R TCs). In certain cases, a compensation cable made up of a less expensive material with similar EMF properties to the TC can be used.

RTDs, on the other hand, can be wired with standard cable for much longer distances; however, they are typically limited by issues related to self-heating errors. In both cases, the extension wiring must be shielded against electrical noise in the plant. Note that TC extension wire is more susceptible to noise than RTD cable.

Sensor construction

A wire-wound RTD consists of fine coiled wire wrapped around a ceramic or glass element, then placed in a sheathed probe for protection (see figure 1). Thin-film RTDs have a platinum coating over a ceramic substrate. Thin-film RTDs are not as susceptible to vibration as wire-

wound RTDs but typically are more limited in their applicable temperature range.

Wire-wound RTDs are typically made from copper, nickel, or platinum alloys. Platinum is the most popular, because it has better accuracy and works over a wider temperature range. RTDs can be taped or cemented to a surface, mounted in a probe, or embedded in a device via threaded holes.

Thermocouples are not as fragile as RTDs, and are typically protected by a metal covering that can be bolted, welded, soldered, or glued to a surface. In some cases, the temperature junction can be left exposed to the environment, allowing for faster response times (see figure 2).

Grounding can be an issue. A TC can build up a static charge that affects its accuracy, so it may need to be grounded. However, TCs can also pick up circuit noise when grounded to an electrical device or machine. Determining whether or not to ground a TC depends on specifics of the particular application, such as the amount of electrical noise present in the measurement environment and the grounding circuit.

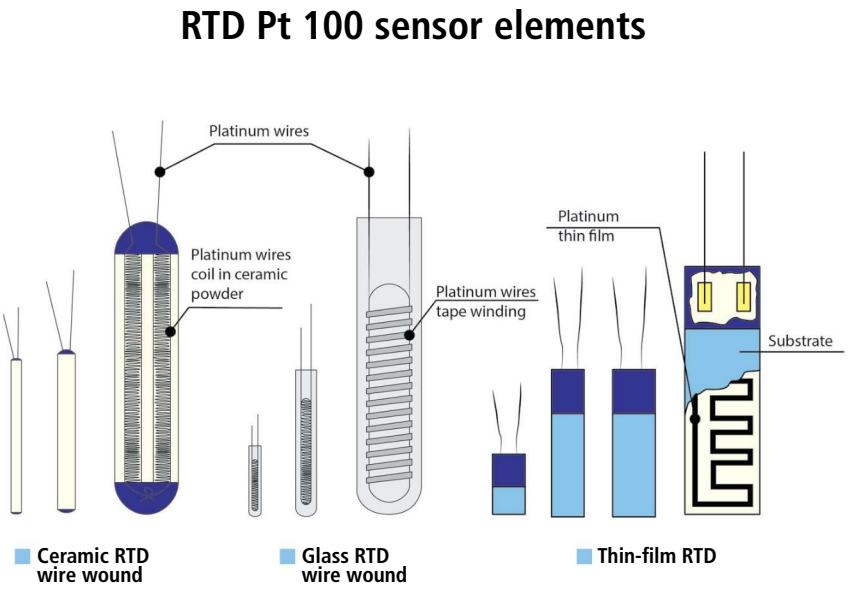


Figure 1. Wire-wound RTD sensing elements are fragile, so they are placed in a protective sheath.

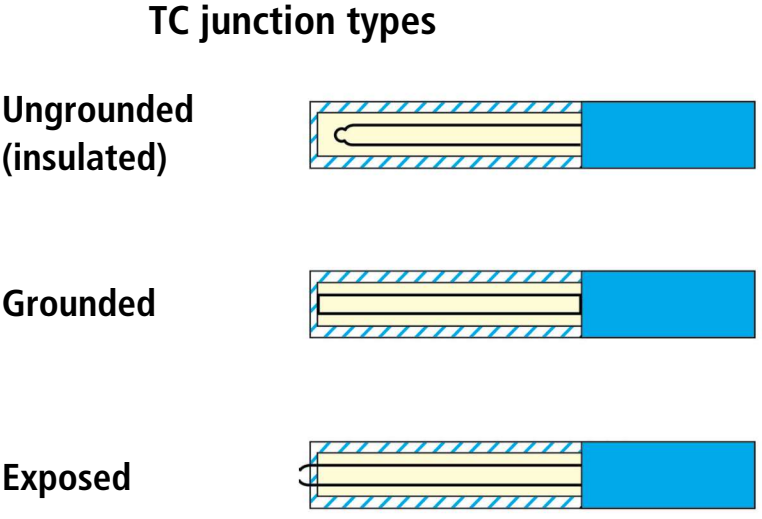


Figure 2. Thermocouple sensors can be grounded or ungrounded, depending on the application.

Nasty environments

Temperature sensing is often done in "unfriendly" environments, such as corrosive, oxidizing, or reducing atmospheres, often accompanied by severe vibration and electrical noise. When selecting an RTD or TC, the environment must be taken into account. If the process environment poses a high degree of risk of sensor failure, a thermowell made of a material that is suitable for the process environment should be considered.

Vibration—Wire-wound RTDs are most susceptible to vibration. Wire-wound RTDs can fail from mechanical stress in high-vibration applications and should not be used. Thin-film RTDs have a higher tolerance for vibration, but they are not as good as TCs, which have the highest resistance to vibration.

Electrical noise—As noted above, the extension wires for both RTDs and TCs are susceptible to electrical noise. In high-noise environments, extension wires should be sheathed, shielded, grounded, and kept as short as possible. An RTD is a better choice in high-noise environments.

Harsh environments—When dealing with a harsh environment, check to see if the RTD or TC sensor manufacturer offers protection against adverse conditions. Wire-wound RTDs in their protective casings are quite rugged and immune to most environmental problems. For additional protection, RTDs can be coated with perfluoroalkoxy (PFA) polytetrafluoroethylene for use in plating baths, highly pressurized systems, or similar applications. RTD extension wires are available with polyvinyl chloride, PFA, or fiberglass insulation for protection.

TCs, especially those with metal cases, are much more rugged than RTDs and better able to deal with corrosive or oxidizing atmospheres. When exposed TC junctions are used, special care must be taken in harsh environments.

TCs are classified by type, i.e., Types E, J, K, N, T, S, R, and B. Each type is suitable for a specific temperature range from –201°C to 1700°C, and the construction of each makes it suitable for use in various environments:

- Type E: suitable for vacuum, inert, mildly oxidizing, or reducing atmospheres
- Type J: may be used, exposed or unexposed, where there is a deficiency of free oxygen
- Type K: typically requires metal or ceramic protection
- Type N: resists oxidation where sulfur is present
- Type T: can be used in either oxidizing or reducing atmospheres
- Types S, R, B: should always be protected with a ceramic tube, a secondary tube of porcelain, and a silicon carbide or metal outer tube as conditions require

RTD – TC comparison

	RTD	TC
Measuring range	<600°C	<600°C
Response time	Fast	Very fast
Dimension	>0.5 mm	<0.5 mm
Vibration	Good	Very good
Self-heating	Little	None
Long-term stability	Very good	Good
Accuracy	Very good	Good
Connection cables	Connection cables	Thermocable
Power supply	Required	Not required

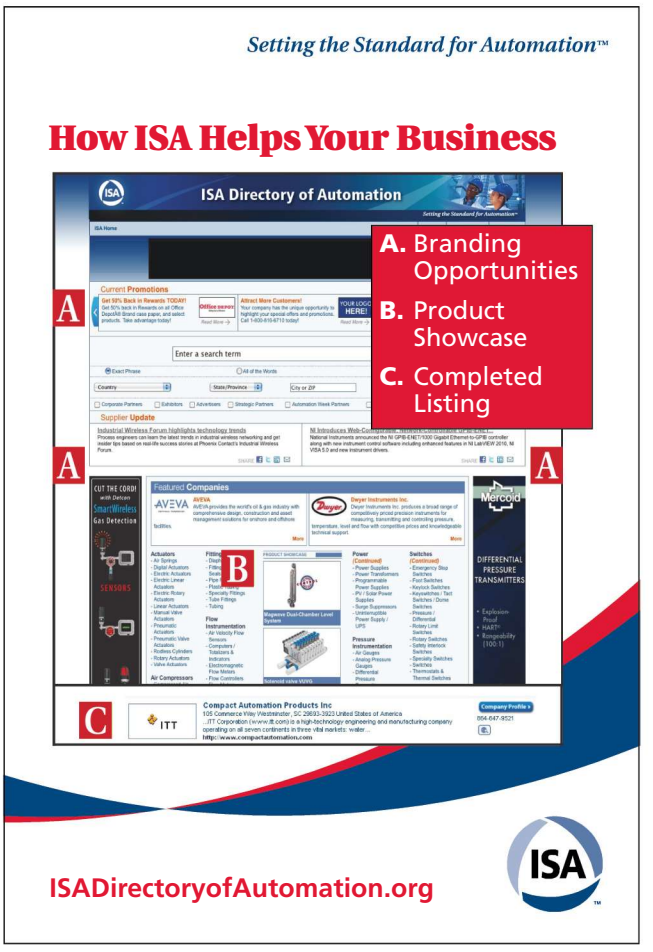
Figure 3. Comparison of thermocouples and RTDs

Other considerations

Figure 3 shows the major considerations involved in selecting a TC or RTD.

Measuring range: RTDs can measure temperatures up to 1000°C, but it can be difficult to get accurate measurements from an RTD at temperatures above 400°C. TCs can measure up to 1700°C. The generally accepted rule is: for temperatures below 850°C, use an RTD for accuracy; for temperatures above 850°C, use a TC. Industrial mea-





surements typically are 200°C to 400°C, so an RTD may be the best choice in that range.

Response time: While both sensors respond quickly to temperature changes, TCs are faster. In certain cases, however, manufacturing processes allow for production of thin-film RTDs with greatly improved response times.

Dimensions: Both sensors are fairly small, at about 0.5 mm diameter. If space is an issue for mounting a sensor, check with the supplier for size and mounting options.

Vibration: In general, TCs are less susceptible to vibration than RTDs. However, in certain manufacturing processes thin-film RTDs can be produced that are capable of much greater vibration resistance than standard RTDs.

Self-heating: An RTD is made of very fine wires or very fine coatings, and requires a voltage from a power supply—TCs do not require power. Although the power required is only 1 mA to 10 mA, it can cause the platinum element in the RTD to "heat up," thus affecting accuracy of the measurement. If long extension wires are used, more power may be needed to overcome resistance in the wires, thus increasing the self-heating problem.

Stability: The long-term stability of an RTD is very good, meaning that its readings will be repeatable and stable for a long time. TCs, on the other hand, tend to drift. The EMF produced by a TC can change over time because of oxidation, corrosion, and other changes in the metallurgical properties of the sensing elements. TC drift is irreversible, and some provision is needed to detect it, such as software or testing for loop resistance.

Accuracy: As a general rule, RTDs are more accurate than TCs. RTDs can produce accuracy of 0.1C, while TCs are typically accurate to only 1C.

Although not a technical issue, TCs are considerably less expensive than RTDs due primarily to lower production costs. Depending on the number of sensors needed for a particular application, this could be a major factor.

Careful selection of the right temperature sensing technology is essential to ensure the best performance, reliability, and cost effectiveness. Suppliers that sell both TCs and RTDs are often good sources of information when deciding which type of sensor to use, as is previous experience with different sensor types in various applications.

To learn more about practical thermocouple thermometry, check out www.isa.org/thermometry.

ABOUT THE AUTHOR

Ehren Kiker (ehren.kiker@us.endress.com) is a product manager with Endress+Hauser with more than 15 years of experience in process control instrumentation.



Technology convergence presents evolving workforce development challenges in combating cyberthreats

By Jay Warmke

Every institution, from the federal government to the corner convenience store, struggles with the ongoing issue of workforce development—and now also with the issue of cybersecurity.

There is the initial struggle to recruit a person who has the interpersonal skills and attitude to succeed, and the knowledge to contribute to the collective effort on day one. And in evolving and technical industries, there is the ongoing task of retraining technicians in emerging technologies, developing job skills from this new knowledge, adapting and applying these skills in the real world, and then evaluating the individual to see if that employee really has absorbed the information now necessary to perform the job.

The challenges of workforce development can clearly be seen in the recent initiative by the National Institute of Standards and Technology (NIST), a part of the U.S. Department of Commerce, as it struggles to address the concerns raised by the growing threat of cyberattacks on our nation's critical infrastructure.

Out of these concerns has emerged the National Initiative for Cybersecurity Education (NICE) Framework, a nationally coordinated effort focused on cybersecurity awareness, education, training, and professional development. Within the NICE Framework, the effort seeks not only to define terms, create job descriptions, and divide security tasks into specialty areas—but it also seeks to come to grips with the areas of education necessary to make the nation's infrastructure more secure.

After those fields of expertise are identified (which is no easy task), how do you go about verifying that an individual is indeed qualified in that specialty? This is where training and third-party certification programs become paramount in preparing a workforce that can identify and alleviate these threats. As part of its process, the NICE initiative is identifying existing certification programs that are applicable and helpful in dealing with and preventing cyberthreats.

As one might assume, most of the currently listed programs were initially designed with network security in mind. Programs such as:

- Global Information Assurance Certification (GIAC) Information Security Fundamentals
- CompTIA Security+
- ISC's Certified Information Systems Security Professional
- Certified Hacking Forensic Investigator

But cybersecurity is not just for computers anymore. The playing field is evolving . . . changing . . . converging.

The threats come from new vulnerabilities

Increasingly, computer networks are moving to wireless platforms. Even our personal cars and public transit systems are incorporating wireless data systems as standard features. With these changes come increased and evolving network vulnerabilities. The world of computing is moving into a landscape previously dominated by those trained in radio frequency (RF) applications.

And at the same time, over the past 10 to 15 years, mobile radio communications and cell and wired telecommunications systems have become more and more Internet Protocol based in an effort to enhance their efficiency, adaptability, and connectivity to existing data systems. The industries are converging.

In the early days of fiber-optic technology, it was touted as the industry's answer to network security. But in recent years, it has been determined that, using only commonly available tools and equipment, fiber-optic cables are as easy to tap as copper—and the security breach is almost undetectable. With millions of miles of fiber transmitting billions of megabytes of data daily—who needs access to the computer hardware?

And how vulnerable are first responders, who have traditionally relied on RF technologies, to cyberthreats? The list of vulnerabilities and technologies affected are seemingly endless.

New specializations will be required

The Electronic Technician Association, International (ETA) has developed more than 80 certification programs focused on skill sets as varied as:

- Certified Network Computer Technician (CNCT)
- Certified Network Systems Technician (CNST)
- Computer Service Technician (CST)
- Wireless Network Technician (WNT)
- Wireless Communications (WCM)
- Communication Site Installer (R56)
- Mobile Communications & Electronics Installer (MCEI)
- Fiber Optics Installer (FOI)
- Fiber Optics Technician-Outside Plant (FOT-OSP)
- Radio Frequency Identification Technical Specialist (RFID)

These programs are widely accepted within their respective industries, and undergo constant review and updates. Many are currently incorporating cyber- and network-security issues into their programs.

Conversely, those interested in cybersecurity issues increasingly need a comprehensive knowledge of diverse technologies—simply to understand those systems and the inherent vulnerabilities they present.

For more information about the certification programs, publications, and technical information available through the ETA, International, visit its website at www.eta-i.org.

ABOUT THE AUTHOR

Jay Warmke is a writer, speaker, and teacher and was recently awarded ETA's 2013 Educator of the Year Award. Warmke also serves on the board of the International Certification Accreditation Council.

ISA-100 wireless standard receives resounding approval in IEC technical voting

ISA-100.11a-2011, *Wireless Systems for Industrial Automation: Process Control and Related Applications,* has received approval from all 21 participating national committees to the International Electrotechnical Commission (IEC) that voted on the IEC version in a recently completed committee draft for vote (CDV) stage.

The CDV stage is the final stage in the IEC process in which technical comments may be considered. The standard, expected to be issued in early 2014 for a final IEC approval stage in which no technical comments may be submitted, will carry the designation IEC 62734 upon publication.

This follows previous approval of ISA-100.11a as an ISA and American National Standard. The latter confirmed that ISA's open consensus standards development procedures as accredited by the American National Standards Institute (ANSI) were followed properly in developing ISA-100.11a. ANSI requires participation not only of suppliers but of all key interest categories.

ISA's ANSI-accredited procedures allow experts from end-user companies to directly participate and vote, ensuring consideration of the views and requirements of those who will actually be deploying wireless systems in real-world industrial applications. That end-user input is now evident in the growing worldwide application of ISA-100.11a technology in processes including oil and gas, liquefied natural gas (LNG), refining, petrochemicals, chemicals, power, biopharmaceuticals, thermal processing, and water treatment.

That end-user input is now evident in the growing worldwide application of ISA-100.11a.

In a presentation at the 2013 ARC Forum, for example, C. B. Chakradhaar of RasGas described the highly cost-effective application of ISA-100.11a-compliant wireless temperature transmitters to monitor a critical component in LNG production. Based on that success, he indicated, ISA-100.11a has been adopted as a plantwide standard for all future wireless applications.

To read about other real-world applications of ISA-100.11a, visit www.isa100wci.org/en-US/Applications/Success-Stories.

The extension of ISA-100.11a to the IEC is the latest in a long and productive collaboration. Many original ISA standards have been adopted to become widely used IEC global standards in vital areas including:

- ISA-84/IEC 61511 Functional Safety
- ISA-88/IEC 61512 Batch Control
- ISA-95/IEC 62264 Enterprise-Control System Integration
- ISA-18/IEC 62682 Management of Alarm Systems
- ISA-99/IEC 62443 Cyber Security for Industrial Automation and Control

2013 ISA Standards & Practices Department awards winners

Each year, the ISA Standards & Practices Department recognizes significant contributions by standards committee members toward achieving major milestones—which typically would result in the publication of an ISA standard, recommended practice, or technical report. Six winners for 2013 were recently announced.

Hal Thomas of exida was recognized for his leadership and technical expertise as ISA84 Working Group 9 chairman in the development of ISA-TR84.00.09-2013, *Security Countermeasures Related to SIS*. The recently approved technical report is expected to be published in early December 2013. It provides guidance on the countermeasures used to reduce the likelihood of a security breach of a safety-instrumented system (SIS) that degrades its ability to perform its function. This relates to cybersecurity from both inside and outside the plant boundary.

The five additional winners of the award were recognized for leadership and technical contributions in developing the recently published ISA-TR106.00.01-2013, *Procedure Automation for Continuous Process Operations – Models and Terminology.* They are:

- Yahya Nazer, Dow Chemical Company, ISA106 co-chair
- William Wray, Bayer MaterialScience, ISA106 co-chair
- Marty King, Chevron Energy Technology Company, ISA106 past co-chair
- David Emerson, Yokogawa, ISA106 lead editor
- Charles Green, Aramco Services Company, ISA106 secretary

ISA–TR106.00.01-2013 addresses good practices for procedure automation in continuous process operations and strategies for incorporating automated procedures into industrial automation and basic process control systems (BPCS). The technical report complements the concepts in key ISA standards that deal with the use of procedures in specific applications, including batch processing (ISA-88) and safety systems (ISA-84). Many such fundamental concepts are common to all procedures without regard to the context in which they are implemented, but there are needs specific to continuous processes. These needs are addressed in a procedure requirements model, procedure implementation model, and physical model as well as in sections concerning mapping of procedures to BPCS components.

For more information about ISA Standards, contact Charley Robinson, crobinson@isa.org, +1 919-990-9213, or visit www.isa.org/standards.

Focus on pressure sensors

Wireless pressure transmitter

Emerson Process Management releases the Rosemount 3051 transmitter for measuring pressure, level, and flow. The device wirelessly monitors assets and has five-year stability and a 10-year power module life. Equipment failure is the cause of nearly half of an operation's downtime, yet many assets go unmonitored due to cost constraints. With the Rosemount 3051 wireless pressure transmitter, users can monitor many assets throughout an operation with a 40–60 percent cost savings over wired installations.



Unmonitored variations cause quality to degrade and throughput to decrease. Increasing the number of measurements provides better insight into the process and helps reduce variation, but the time and cost to add a new measurement can be a barrier. These devices allow operations to install additional measurement points quickly and economically, providing the precise and stable measurements needed to reduce variability and maximize throughput.

With the Rosemount 3051 wireless pressure transmitter, users can monitor flow and pressure in compressed air, steam and water systems to benchmark energy usage, identify energy savings opportunities throughout an operation, and provide energy management and accurate internal billing.

Emerson Process Management, www.emersonprocess.com

Digital pressure sensors

The Honeywell model DPS digital pressure sensors with CANopen are highly configurable, so customers can choose from multiple pressure types, accuracy levels, pressure ranges, pressure connections, and electrical terminations to meet specific application needs. Configurations for digital measurements are fully temperature compensated and calibrated for pressure ranges from 10 psi to 10 K psi or 1 bar to 700 bar or 70 kPa to 70,000 kPa, with accuracy ranges of 0.25 percent and 0.1 percent. Operating and storage temperature ranges from –25°C to 85°C (–13°F to 185°F). Built to withstand harsh environments, the protocol communication allows customers to connect to longer cable distances without sacrificing accuracy. Customers can configure the devices to meet their specific application needs due to a wide pressure range, multiple pressure engineering units that eliminate having to make mathematical conversions, and a choice of connectors that simplify use.



Honeywell, www.honeywell.com

EJA series pressure transmitters

Yokogawa Corporation of America released the enhanced version of the DPharp EJA series pressure/differential pressure transmitters. While remaining in the same price range, the enhanced series now offers improved performance and functionality. As standard features, the devices have the same high-speed response and multisensing functionality as the high-end DPharp EJX series.

Pressure/differential pressure transmitters are typically used in the oil, petrochemical, and chemical industries to measure liquid, gas, and steam flow rates as well as liquid levels in tanks. The DPharp EJA series transmitters are compact, multipurpose sensors with a large installed base. The transmitters simultaneously measure differential and static pressure. A single enhanced EJA transmitter can measure both the liquid level and the pressure in a tank. Previously, this could only be done with two EJA transmitters. The response time is now 90 ms, which equals the capability of the EJX series transmitters. Specifically, this is the time from the detection of pressure to the output of an electric signal. The improved transmitters weigh 30 percent less and have indicators that are nearly twice the size of those on the nonenhanced version. Additionally, the optional ultra-low copper aluminum housing can provide much improved corrosion resistance over standard aluminum housings, making it a suitable alternative to 316SS for offshore environments.



The transmitters now comply with the IEC 61508 international safety standard on the functional safety of electrical/electronic devices and are certified for use in safety integrity level 2 applications as stipulated by the IEC.

Yokogawa Corporation of America, www.yokogawa.com

Model SP007 differential pressure transmitter

The Tecsis Model SP007 differential pressure transmitter has a bidirectional wet/wet design that uses silicon technology for reliable data in tank level measurement. It monitors flow rate in heat exchangers or filters and in a variety of test stand applications. The SP007 is built to perform in the most demanding conditions with welded stainless-steel construction and a 3x-safe overload rating. This unit is available in ranges from 0–1 to 0–500 psid with standard accuracy of 0.25 percent, or optional high accuracy of 0.1 percent.

Amplified output of 0–5 VDC, 0–10 VDC or 4–20 mA is offered, and there are five electrical connections to choose from. With six different pressure ports available, the SP007 will easily integrate into most any system. Operating temperature is –40°F to 240°F, compensated from 40°F to 140°F.

Tecsis, www.tecsis.com

Dry block calibrator



The Intempco DB100 Dry Block is a miniature temperature calibrator that is lightweight and portable. The unit measures 6.25 in by 5 in by 2 in. The dry block can be ordered with various well sizes to ensure an optimal fit between the sensor and the controlled thermal mass as well as to suit specific needs. It is factory calibrated using an advanced fuzzy logic controller and has a four-digit LED display; the unit can be ordered to indicate in either Fahrenheit or Celsius. The calibration range is from ambient temperature plus 10°C to 300°C (86°F to 570°F).

Intempco, www.intempco.com

PX409-USBH pressure transducer



Omega's PX409-USBH series of high-speed pressure transducers connect directly to a computer and feature long-term stability, 316L SS wetted parts, and 1000/reading/second capacity. They have secondary containment. The micro-machined silicon design is suitable for pressure or level applications in laboratory, test platforms, or biopharmaceutical applications that require a rugged, high-accuracy transducer.

Omega Engineering, www.omega.com

Explosion-proof pressure switch



The Ashcroft A-Series miniature pressure switch is available in an SIL-capable explosion-proof configuration for hazardous areas. With 316 stainless-steel construction, an IP67-rated enclosure, and an operating temperature range from –40°C to 89°C (–40°F to 192°F), the dual-seal rated pressure switch stands up to the most extreme environmental conditions. Agency approvals and certifications include FM, UL, ATEX, IECEx, CSA, CRN, CE, RoHS as well as SIL 3 capability. Pressure ranges from vacuum through 7500 psi are offered, along with a broad selection of pressure inlet fittings.

Ashcroft, www.ashcroft.com

ProSense MPS25 pressure switches



AutomationDirect introduces the ProSense MPS25 series mechanical pressure switches. These switches have an all-welded 316 stainless-steel sealed diaphragm actuator design (for up to 100 psi) or a direct-acting 316 stainless-steel piston design with a Buna-N O-ring (for 200 to 7500 psi). The 316 stainless-steel housing resists vibration and shock while providing reliable operation over a wide operating temperature range. Pressure ranges from vacuum to 7500 psig are available; all models include a ¼-inch NPT process connection and a precision snap-acting SPDT, 3 A, mechanically operated switch output.

The mechanical pressure switches are available with an integral 6-foot cable with ½-inch NPT male conduit connector, or a DIN 175301-803C L-connector. The switches are UL, CSA, CE and RoHS compliant and have a three-year warranty.

AutomationDirect, www.automationdirect.com

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Workforce sustainability in the process industry: Is there an experience gap?

By Tom O'Banion



During a recent customer advisory board meeting at Emerson's Micro Motion, one of the top three issues brought up by participants from the food, beverage, chemical, oil and gas, and refining industries was workforce sustainability. When we asked for more detail, the advisory board members specifically voiced concerns that as older workers retire, tribal knowledge is getting lost. It is not being transferred to the new and younger replacements. The new hires are unaware of measurement instrumentation capabilities and how to maintain or troubleshoot the equipment in their plants. Due to this knowledge gap, companies in the process industry are asking for more training from their instrument vendors. This training is being delivered in multiple ways via on-site, factory, and web-based venues.

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Process industries in the U.S. have seen a decline in total employment during the past 20 years. For instance, the approximate number of total employees in the U.S. chemical sector fell from about 923,000 to about 790,000 between January 2003 and January 2013 (U.S. Department of Labor, www.isa.org/link/USDOL). During this period, production volume doubled—and productivity and global competitiveness increased—which was all good for the industry. However, after two decades with little or no hiring and workforce attrition, an unanticipated experience gap developed. With the lack of hiring, some companies who were previously famous for their training and development greatly reduced or eliminated their technical training for new employees.

More recently, we have seen a substantial number of retirements in these industries and an increase in hiring. This experience gap has driven a renewed focus on training and development and caused companies to rely on their suppliers for assistance.

We are also starting to see that our customer's equipment is aging. Much of their instrumentation and control equipment was purchased years ago. The newer employees—while on average better educated—are much less familiar with these legacy devices.

This aging workforce is not just a problem for the process industry; this is a trend that is starting to play out across many industries and markets. It is important to embrace this change and work together to find ways to fix it. Training is a great way to overcome this phenomenon. Companies with employees who need development and training should closely align with vendors and groups, such as local colleges and universities. If they can work together to identify the training needs and develop a curriculum that combines online and classroom training, everyone wins.



Source: Bureau of Labor Statistics

ABOUT THE AUTHOR

Tom O'Banion (tom.obanion@emerson.com) is the director of global chemical industry marketing for Micro Motion, and has been with them since 1987 in a variety of marketing and sales roles. These roles include greenhouse gas compliance, process and natural gas marketing development, and technical sales management covering Canada, Taiwan, and mainland China. He holds a B.S. degree in chemical engineering degree from the University of Colorado and an MBA from the University of Denver. Previous employers include Conoco Oil and MW Kellogg (now KBR). He is a "retired" triathlete and competitive runner.

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